



IT3021 – Data Warehouse & Business Intelligence

BSc(Hons) in Information Technology

Specialized in Data Science

3<sup>rd</sup> Year

Faculty of Computing

SLIIT

## **Assignment 2**

**Name : B I V Perera**

**IT Number : IT23725010**

## Step 01: Data Source Description

The data source for Assignment 2 is the ToyStore\_DWH data warehouse implemented and loaded in Assignment 1. It is a Star Schema data warehouse built on SQL Server containing 1 fact table and 6 dimension tables, loaded with Toy Store E-Commerce transactional data spanning from March 2012 to March 2015. The data warehouse was designed following the Kimball methodology with a single fact table (FactSales) at the centre, surrounded by six dimension tables that provide analytical context.

### 1.1 Star Schema :

The following star schema diagram illustrates the structure of the ToyStore\_DWH data warehouse, showing the central fact table FactSales and its relationships with the six surrounding dimension tables.

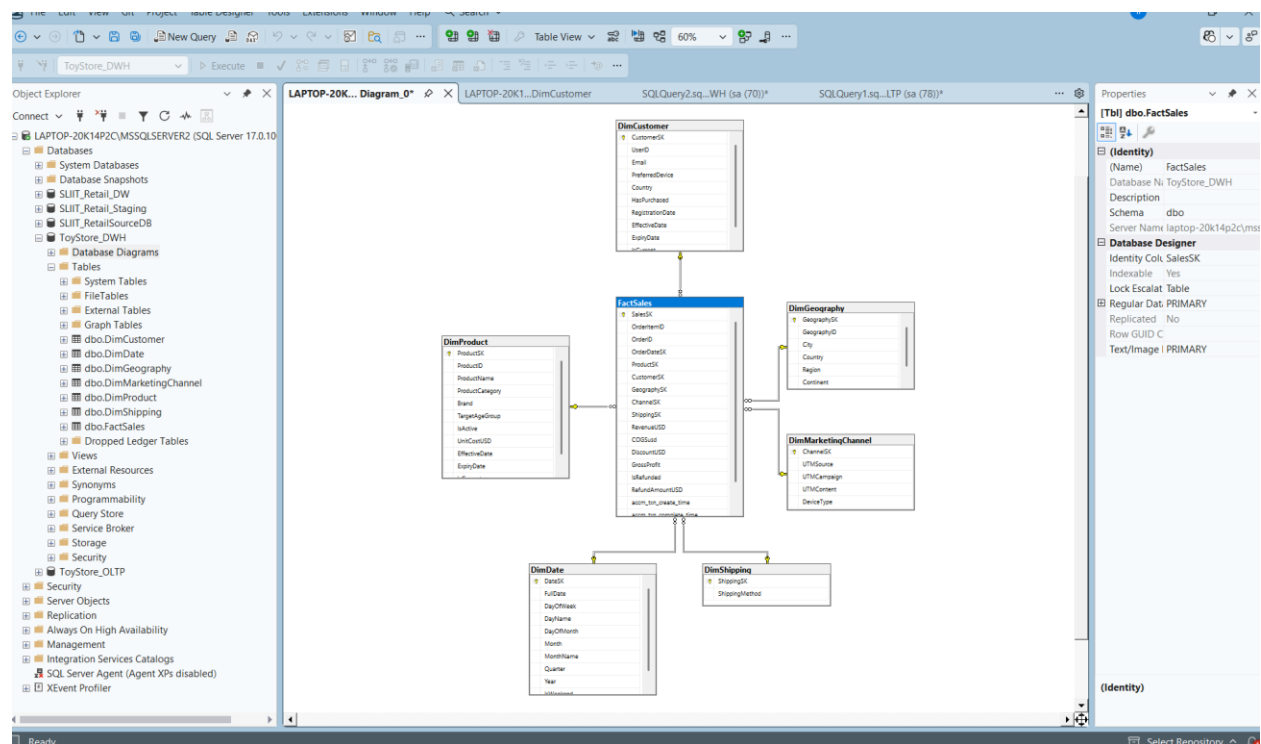


Figure 1: Star Schema Diagram of ToyStore\_DWH showing FactSales connected to 6 dimension tables

### 1.2 Table Descriptions :

**FactSales** - The central fact table containing one row per order line item. It stores all quantitative measures and foreign keys linking to all 6 dimension tables.

**DimDate** - Time dimension generated using a SQL WHILE loop covering all dates from 2012-01-01 to 2015-12-31. Supports time-based analysis at day, month, quarter and year levels.

DimProduct - Product dimension loaded from the enriched products.xlsx Excel file. Implements SCD Type 2 to track historical changes in product attributes.

DimCustomer - Customer dimension loaded from users.csv. Implements SCD Type 2 to track historical changes in PreferredDevice and Country attributes.

DimGeography - Geography dimension loaded from the enriched geography.xlsx Excel file. Covers 10 countries across multiple regions and continents.

DimMarketingChannel - Marketing channel dimension loaded from website\_sessions\_clean.csv. Contains 13 distinct combinations of UTM marketing attributes and device type.

DimShipping - Shipping dimension containing the 4 distinct shipping methods used in the Toy Store orders.

| Table               | Type               | Rows    | Source          |
|---------------------|--------------------|---------|-----------------|
| FactSales           | Fact Table         | 40,025  | SQL Server OLTP |
| DimDate             | Standard Dimension | 1461    | Generated SQL   |
| DimProduct          | SCD Type 2         | 4       | Excel           |
| DimCustomer         | SCD Type 2         | 394,318 | CSV             |
| DimGeography        | Standard Dimension | 18      | Excel           |
| DimMarketingChannel | Standard Dimension | 13      | CSV             |
| DimShipping         | Standard Dimension | 4       | Derived         |

The screenshot displays the SQL Server Enterprise Manager interface. The central pane shows a SQL query executed in the 'ToyStore\_DWH' database. The query is as follows:

```
1 USE ToyStore_DWH;
2
3 SELECT
4     t.name AS TableName,
5     SUM(p.row_count) AS TotalRows
6 FROM sys.tables t
7 JOIN sys.dm_partition_stats p
8     ON t.object_id = p.object_id
9 WHERE p.index_id IN (0,1)
10 GROUP BY t.name
11 ORDER BY t.name;
```

The bottom pane shows the results of the query, which is a table with two columns: 'TableName' and 'TotalRows'. The results are as follows:

| TableName                  | TotalRows |
|----------------------------|-----------|
| DimCustomer                | 394318    |
| DimDate                    | 1461      |
| DimGeography               | 18        |
| DimMarketingChannel        | 13        |
| DimProduct                 | 4         |
| DimShipping                | 4         |
| FactSales                  | 40025     |
| FactSales_ErrorLog         | 0         |
| Stage_OrderCompletions     | 100       |
| Stage_UnmatchedCompletions | 0         |
| syndiagrams                | 1         |

The right-hand pane shows the 'Properties' window for the current connection, displaying connection parameters and details.

**Aggregate Status**

- Connection: 1
- Elapsed time: 00:00:00.548
- Finish time: 3/23/2026 3:20:36
- Name: LAPTOP-20K14P2i
- Rows returned: 11
- Start time: 3/23/2026 3:20:35
- State: Open

**Connection**

- Connection: LAPTOP-20K14P2i

**Connection Details**

- Connection: 00:00:00.548
- Connection: Not encrypted
- Connection: 3/23/2026 3:20:36
- Connection: 11
- Connection: 3/23/2026 3:20:35
- Connection: Open
- Display name: LAPTOP-20K14P2i
- Login name: sa
- Server name: LAPTOP-20K14P2i
- Server version: 17.0.1000
- Session trace: 77
- SPID: 77
- TDS protocol: 0x74000004

**Name**

The name of the connection.

Figure 2: ER Diagram showing table relationships and key columns

## Step 02: SSAS Cube Implementation

This step describes the process of creating an SSAS (SQL Server Analysis Services) Multidimensional cube using the ToyStore\_DWH data warehouse as the data source. The cube was created in Visual Studio 2022 using the Analysis Services Multidimensional project template.

### 2.1 Create Data Source:

The first step in creating the SSAS cube is to establish a connection to the ToyStore\_DWH data warehouse. A new Data Source was created in Visual Studio pointing to the SQL Server instance LAPTOP-20K14P2C\MSSQLSERVER3 using Windows Authentication with the service account for impersonation. This ensures secure and reliable connectivity between the SSAS cube and the underlying data warehouse.

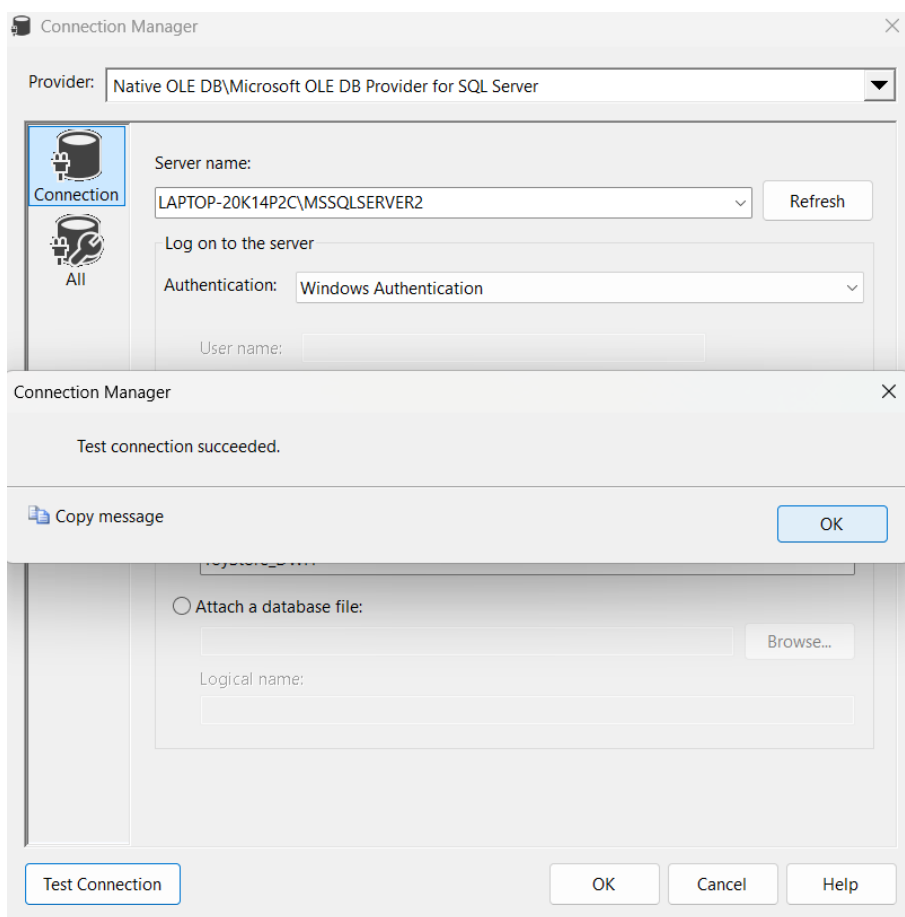


Figure 3: Data Source Designer showing connection to ToyStore\_DWH using Windows Authentication

## 2.2 Create Data Source Views:

After creating the data source, a Data Source View (DSV) named `ToyStore_DSV` was created. The DSV includes all 7 tables from the `ToyStore_DWH` database: `FactSales`, `DimDate`, `DimProduct`, `DimCustomer`, `DimGeography`, `DimMarketingChannel`, and `DimShipping`. The DSV provides a logical view of the data and shows the relationships between tables.

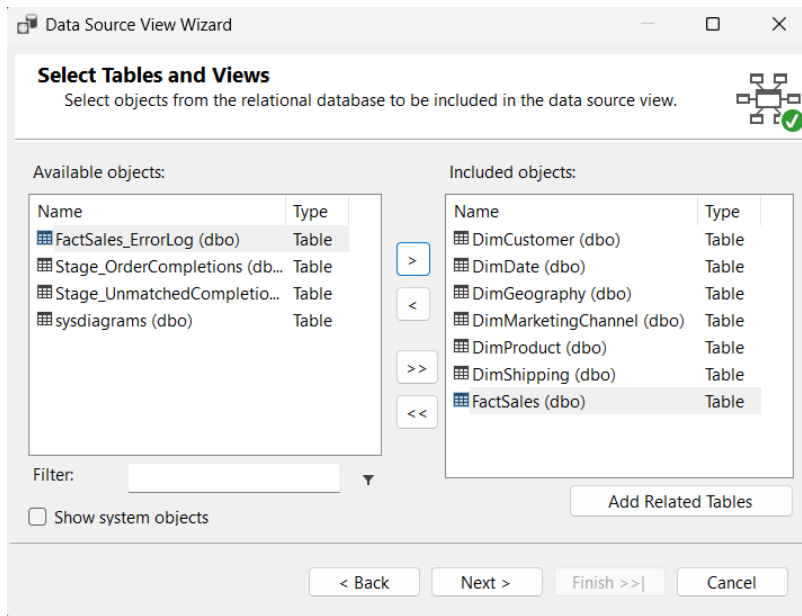


Figure 4: Data Source View wizard showing table selection

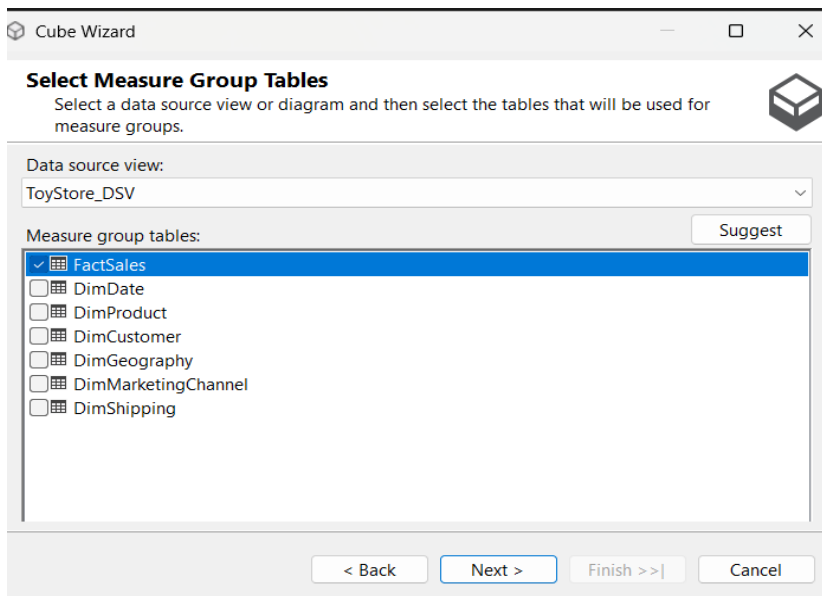


Figure 5: Data Source View configuration

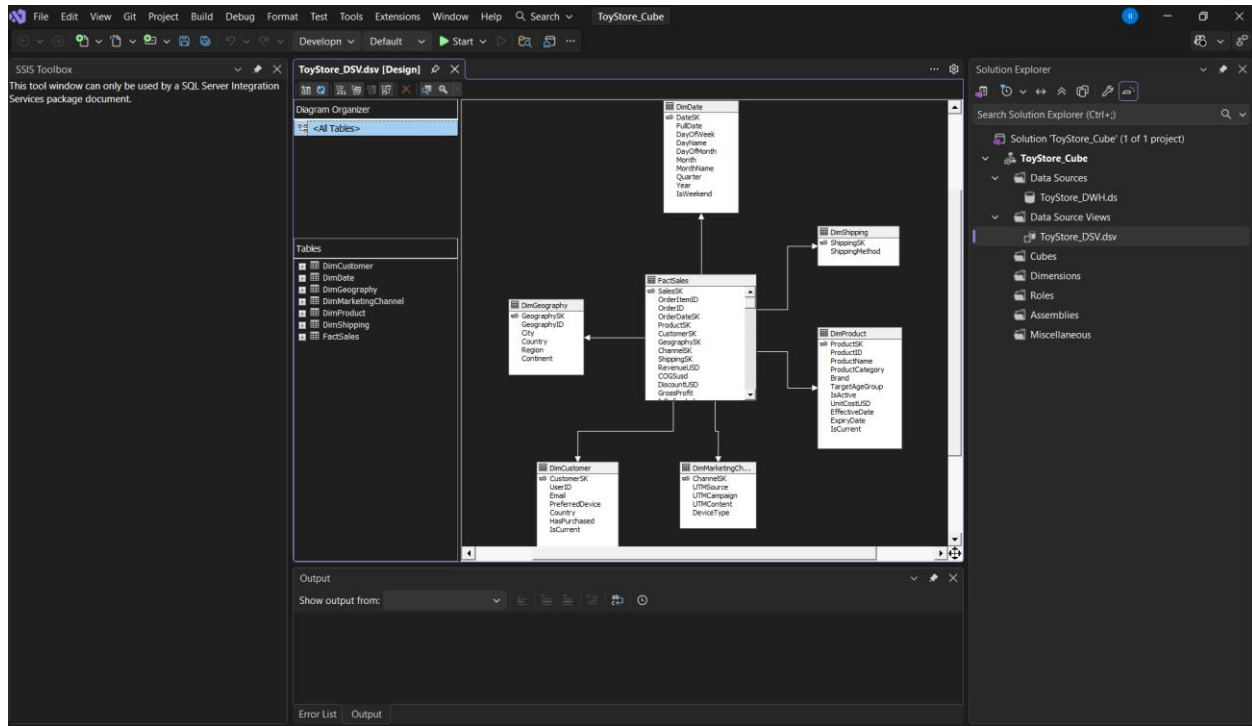


Figure 6: Completed Data Source View showing all 7 tables connected with relationship lines

### 2.3 Create Cubes:

The SSAS cube named ToyStore\_Sales\_Cube was created using the Cube Wizard in Visual Studio. The FactSales table was selected as the measure group table. All available measures were included: RevenueUSD, COGSUSD, DiscountUSD, GrossProfit, RefundAmountUSD, and IsRefunded. All 6 dimension tables were connected to the cube to enable multi-dimensional analysis.

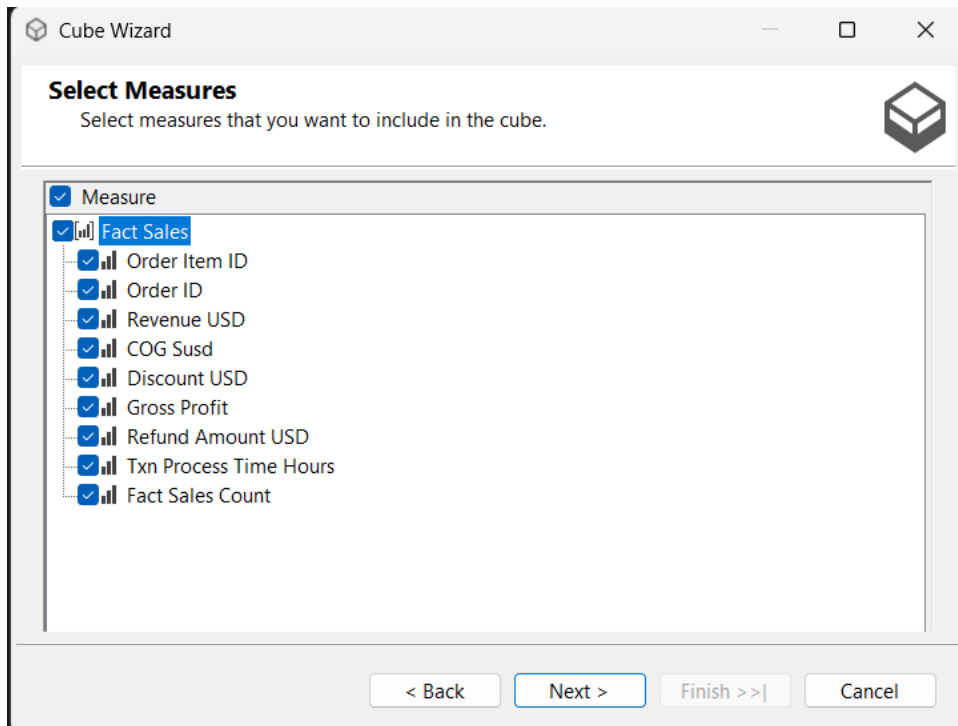


Figure 7: Cube wizard showing measure selection from FactSales

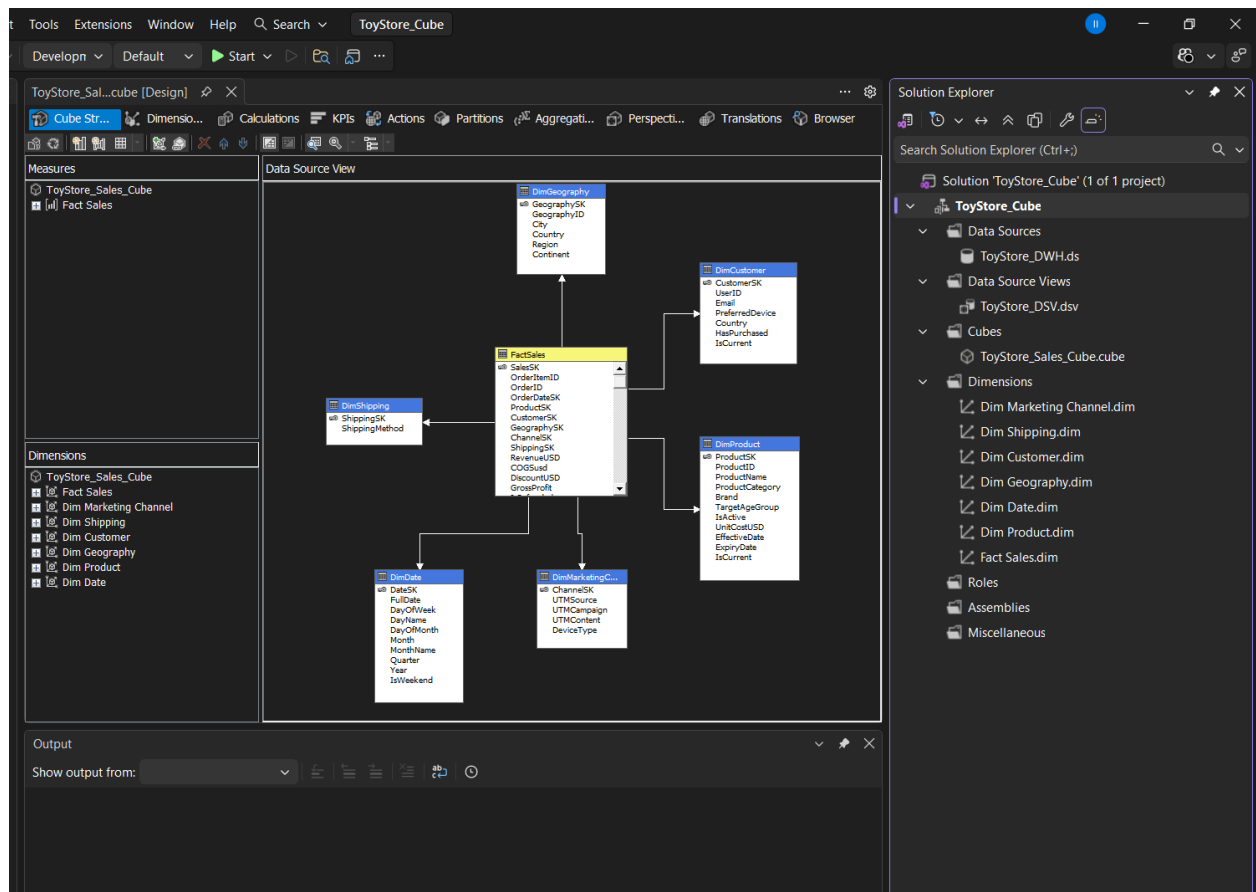


Figure 8: Cube structure showing all measures and all 6 connected dimensions



## 2.4 Add Hierarchies

Three user-defined hierarchies were created in the cube to enable hierarchical navigation and OLAP operations. Hierarchies allow users to drill down from high-level summaries to detailed data, which is essential for meaningful business analysis.

### 1. Geography Hierarchy

The Geography Hierarchy was created in the DimGeography dimension with four levels: Continent → Region → Country → City. This hierarchy enables geographic drill-down analysis, allowing users to start at the continental level and navigate down to individual cities.

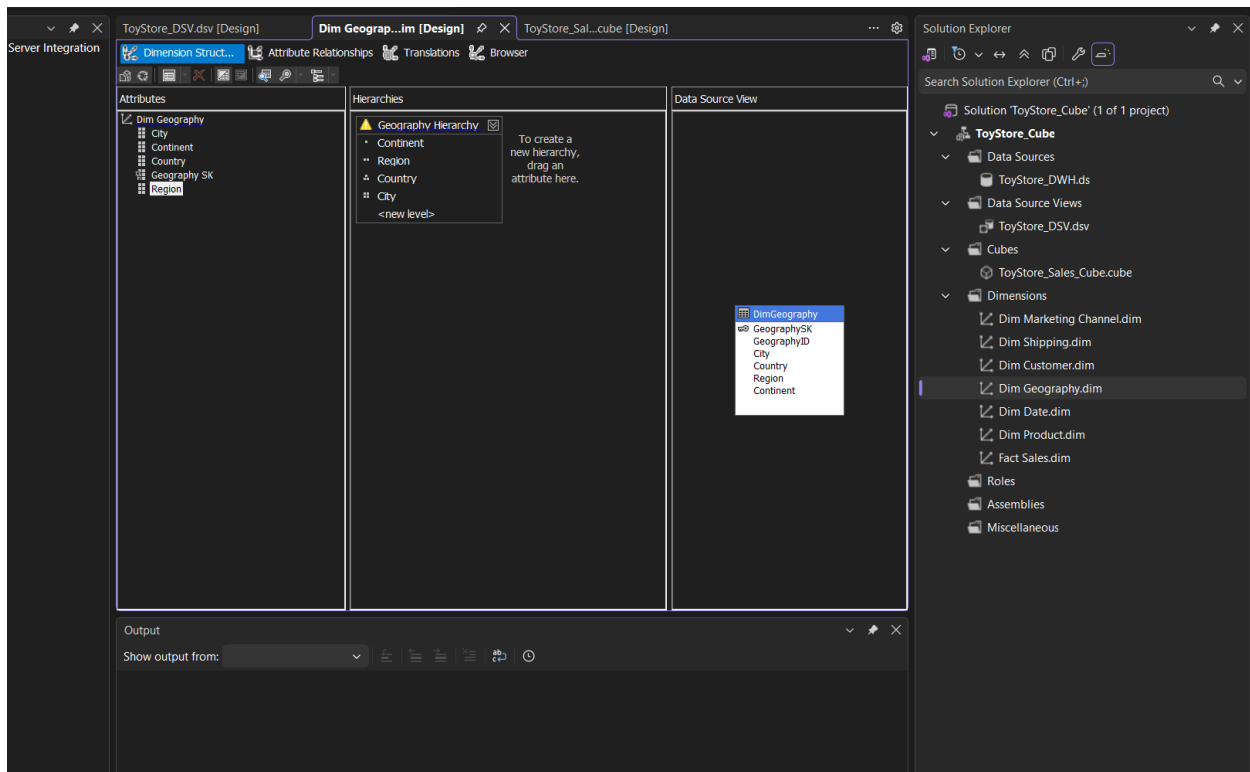


Figure 9: Geography Hierarchy showing 4 levels: Continent → Region → Country → City

## 2. Date Hierarchy :

The Date Hierarchy was created in the DimDate dimension with four levels: Year → Quarter → Month → Day Of Month. This hierarchy enables time-based drill-down analysis, which is the foundation for Roll-Up and Drill-Down OLAP operations demonstrated in Step 3.

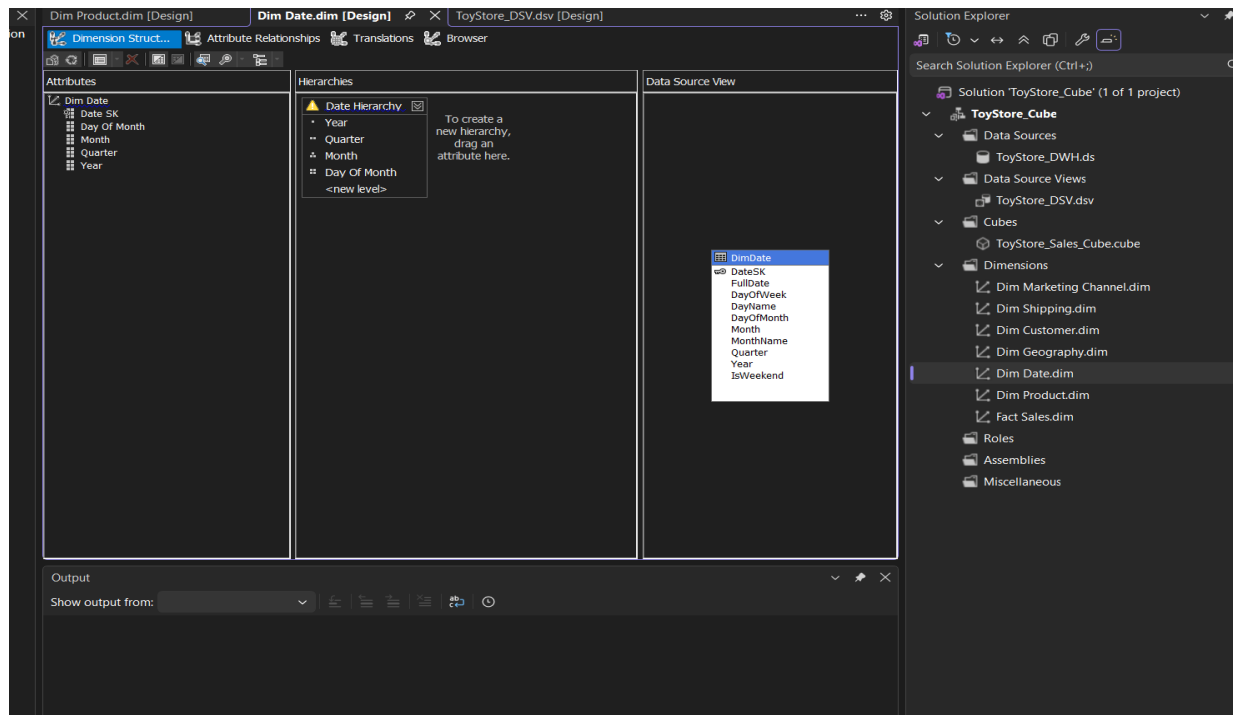


Figure 10: Date Hierarchy showing 4 levels: Year → Quarter → Month → Day Of Month

### 3. Product Hierarchy:

The Product Hierarchy was created in the DimProduct dimension with three levels: Product Category → Brand → Product Name. This hierarchy enables product-based analysis, allowing users to analyse revenue at category level and drill down to specific products.

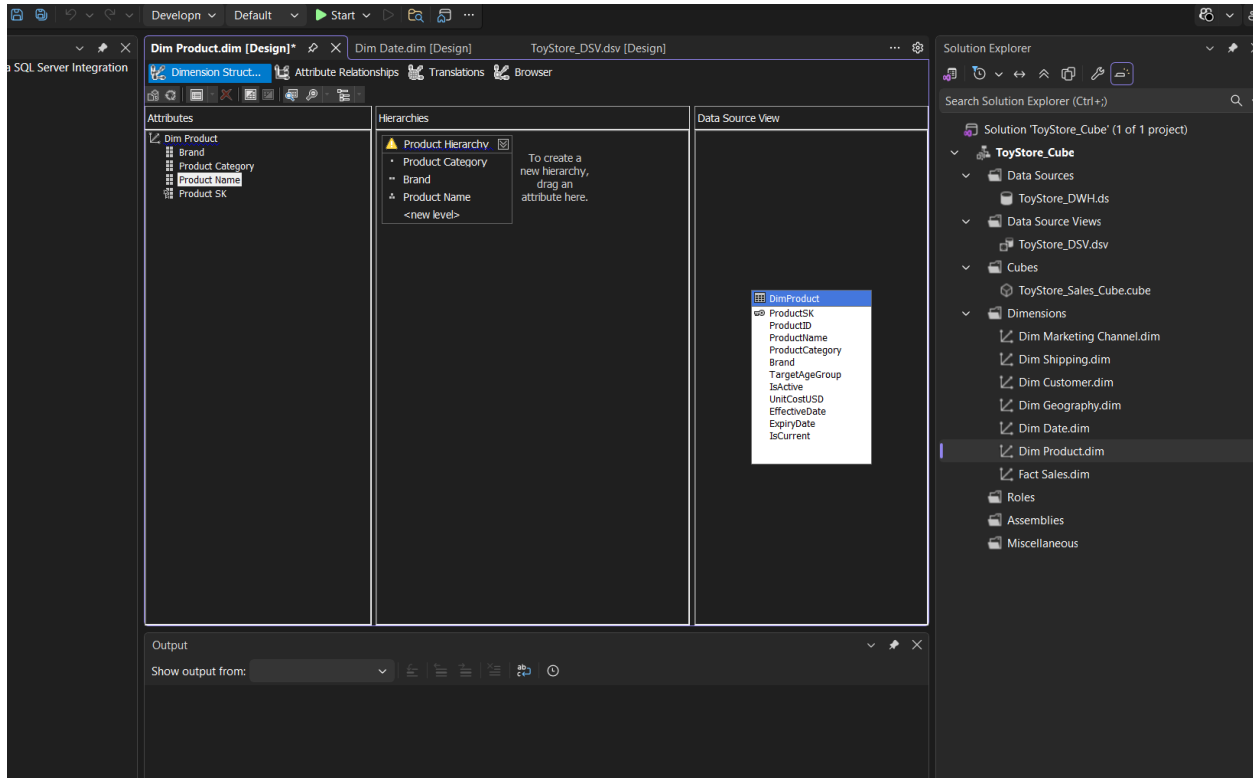


Figure 11: Product Hierarchy showing 3 levels: Product Category → Brand → Product Name

## 2.5 Deploy Cube:

After completing the cube design including all measures, dimensions and hierarchies, the cube was deployed to the SQL Server Analysis Services instance at LAPTOP-20K14P2C\MSSQLSERVER3. The deployment was successful as confirmed by the Output window showing Deploy: 1 succeeded, 0 failed.

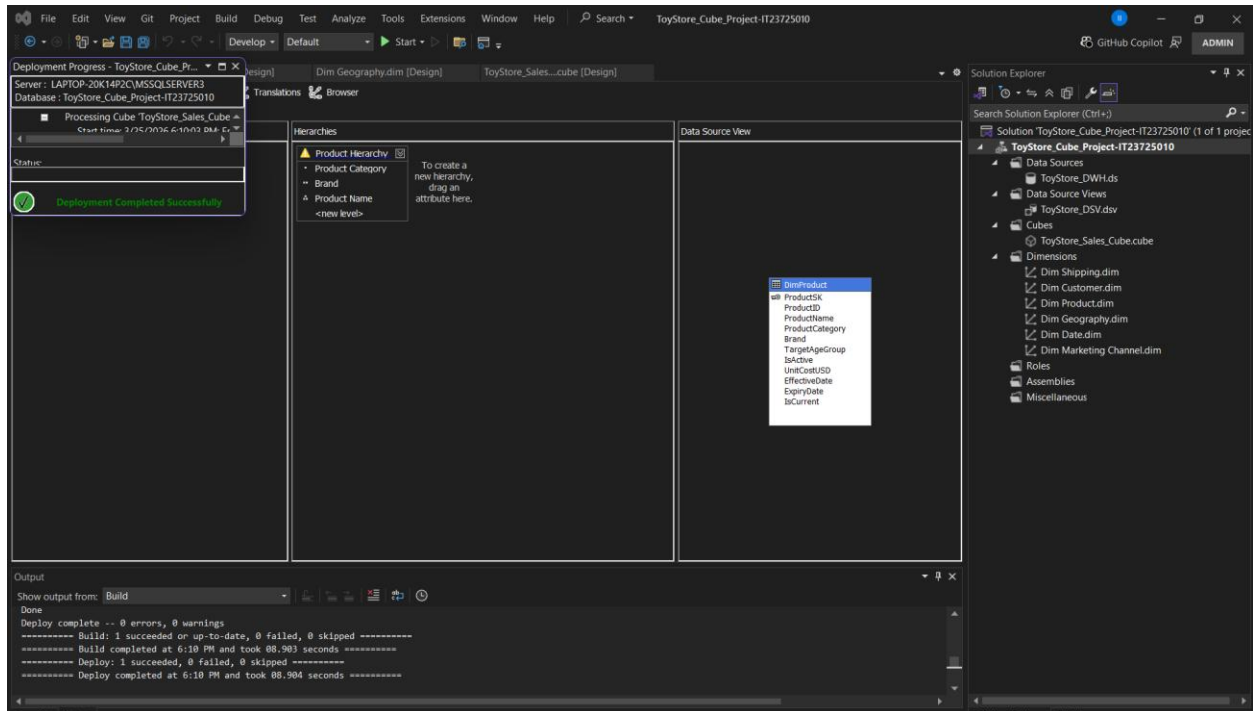


Figure 12: Visual Studio Output window showing successful cube deployment

## 2.6 Browse the Cube:

After successful deployment, the cube was browsed in Visual Studio to verify that it is working correctly. The Browser tab was used to drag measures and dimension attributes to view the data. Revenue USD was displayed by Year, confirming that the cube processes and returns data correctly from the ToyStore\_DWH data warehouse.

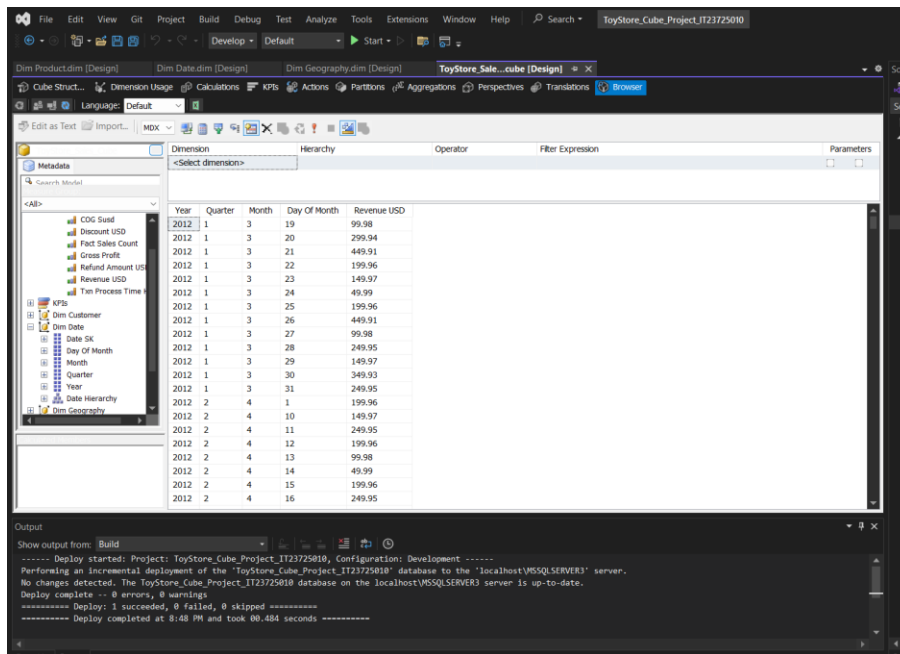


Figure 13: Cube Browser showing Revenue USD by Year, confirming successful cube deployment

## Step 03: Demonstration of OLAP Operations

This step demonstrates all five OLAP operations using Microsoft Excel connected to the ToyStore\_Sales\_Cube via an Analysis Services connection. PivotTables were used to demonstrate each operation, with each operation saved on a separate sheet in the Excel workbook.

### 3.1 Connect Excel to SAAS cube

Excel was connected to the SSAS cube using the Data tab → Get Data → From Database → From Analysis Services. The server name LAPTOP-20K14P2C\MSSQLSERVER3 was used to establish the connection to the ToyStore\_Cube\_Project-IT23725010 database and the ToyStore\_Cube\_Sales cube.

Data Connection Wizard

### Connect to Database Server

Enter the information required to connect to the database server.

1. Server name: localhost\MSSQLSERVER3

2. Log on credentials

☒ Use Windows Authentication

☐ Use the following User Name and Password

User Name:

Password:

Cancel < Back Next > Finish

Figure 14: Excel connected to SSAS cube via Analysis Services connection

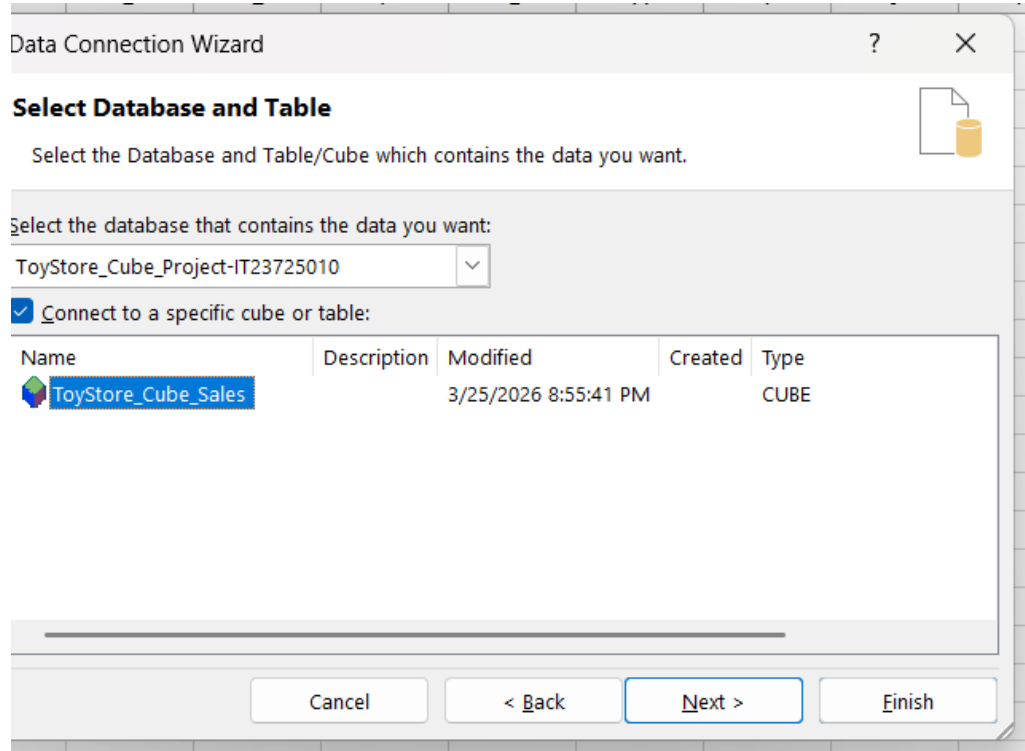


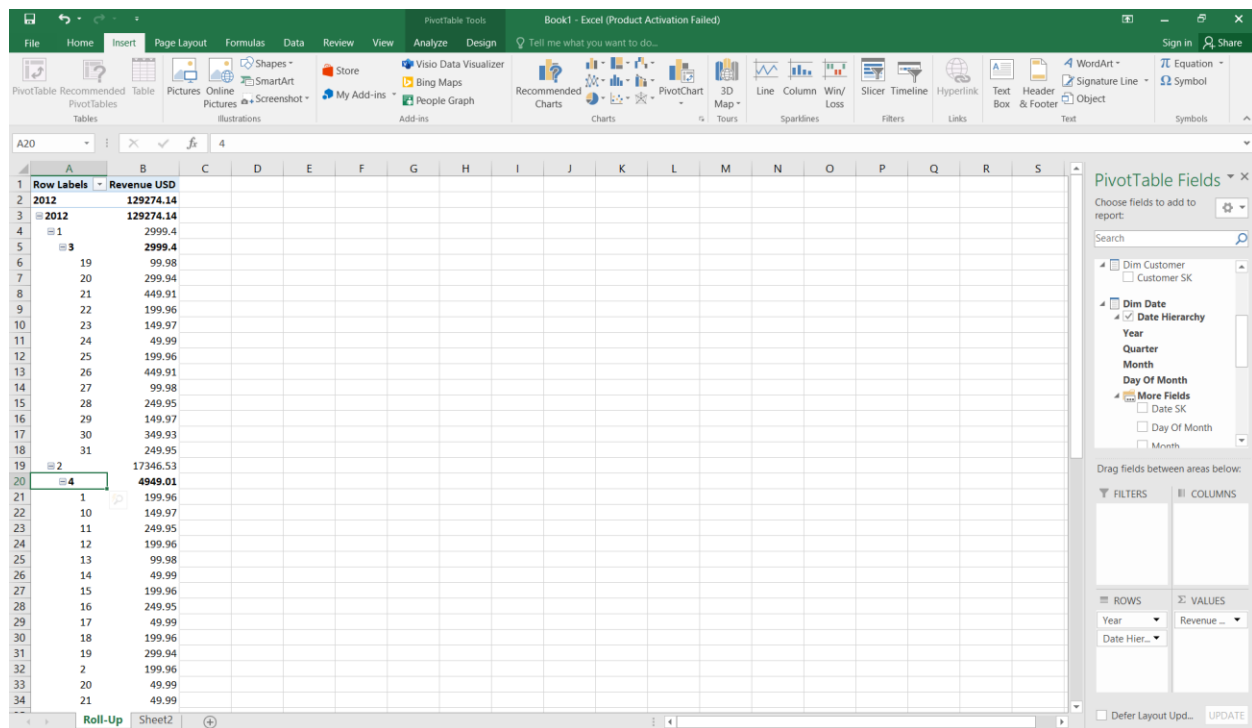
Figure 15: PivotTable Fields pane showing all cube dimensions and measures available in Excel

### 3.1 Operation 1 – Roll- up

Roll-Up is an OLAP operation that aggregates data from a lower level of granularity to a higher level. It moves up the dimension hierarchy, summarising detailed data into broader categories. The Roll-Up operation was demonstrated using the Date Hierarchy, showing how individual daily/monthly revenue can be progressively aggregated up to quarterly and then annual totals.

The same PivotTable is shown at four levels of aggregation. First at Day of Month level showing individual daily revenue, then rolled up to Month level showing monthly revenue totals, then further rolled up to Quarter level showing quarterly aggregates, and finally fully rolled up to Year level showing total annual revenue. This demonstrates how data can be summarised progressively up the hierarchy.

Date of Month -> Month -> Quarter -> Year





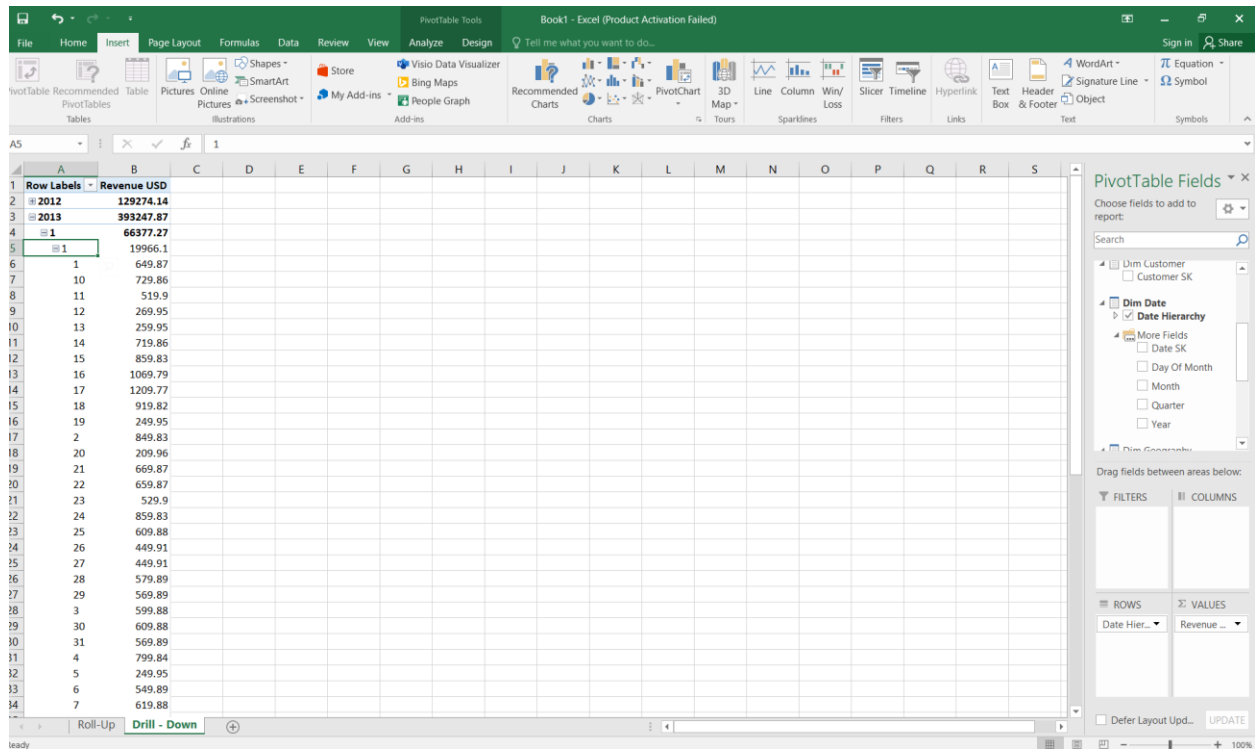


Figure 17: Drill-Down operation — Expanding from Year 2013 to show quarterly, monthly and daily revenue breakdown

### 3.4 Operation 3 – Slice

Slice is an OLAP operation that filters the data cube by fixing one dimension to a specific single value, effectively 'slicing' through the cube to show a subset of data. This operation reduces the number of dimensions being viewed while keeping all other dimensions intact.

A PivotTable was created showing Revenue USD by Year. A Country slicer was added from the DimGeography dimension. Selecting 'United States' in the slicer filtered the entire PivotTable to show only revenue from United States orders, filtering out all other countries. This demonstrates how a single dimension value filters the entire dataset, creating a two-dimensional slice of the cube.

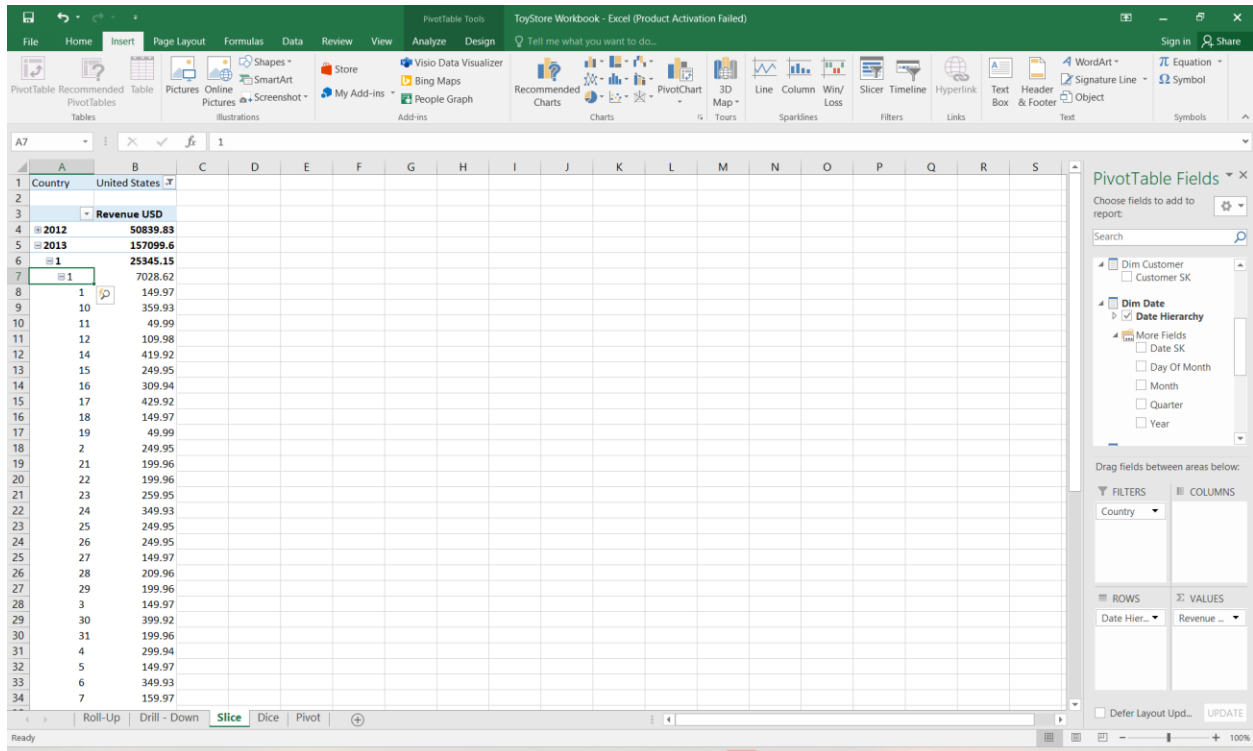


Figure 18: Slice operation — Data sliced to show only United States orders filtered by Country slicer

### 3.5 Operation 4 – Dice

Dice is an OLAP operation that filters the data cube by multiple dimension values simultaneously, creating a sub-cube from the original data. Unlike Slice which fixes one dimension, Dice applies filters across multiple dimensions at the same time, narrowing down the data from multiple angles.

A PivotTable was created showing Revenue USD by Year with Product Category as columns. Two slicers were added simultaneously: a Country slicer (selecting United States and Canada) and a Region slicer (selecting North America and South America). Both slicers filter the PivotTable simultaneously, creating a sub-cube that shows only revenue for selected countries AND selected regions at the same time. This demonstrates how multiple dimension filters combine to create a more targeted data view.

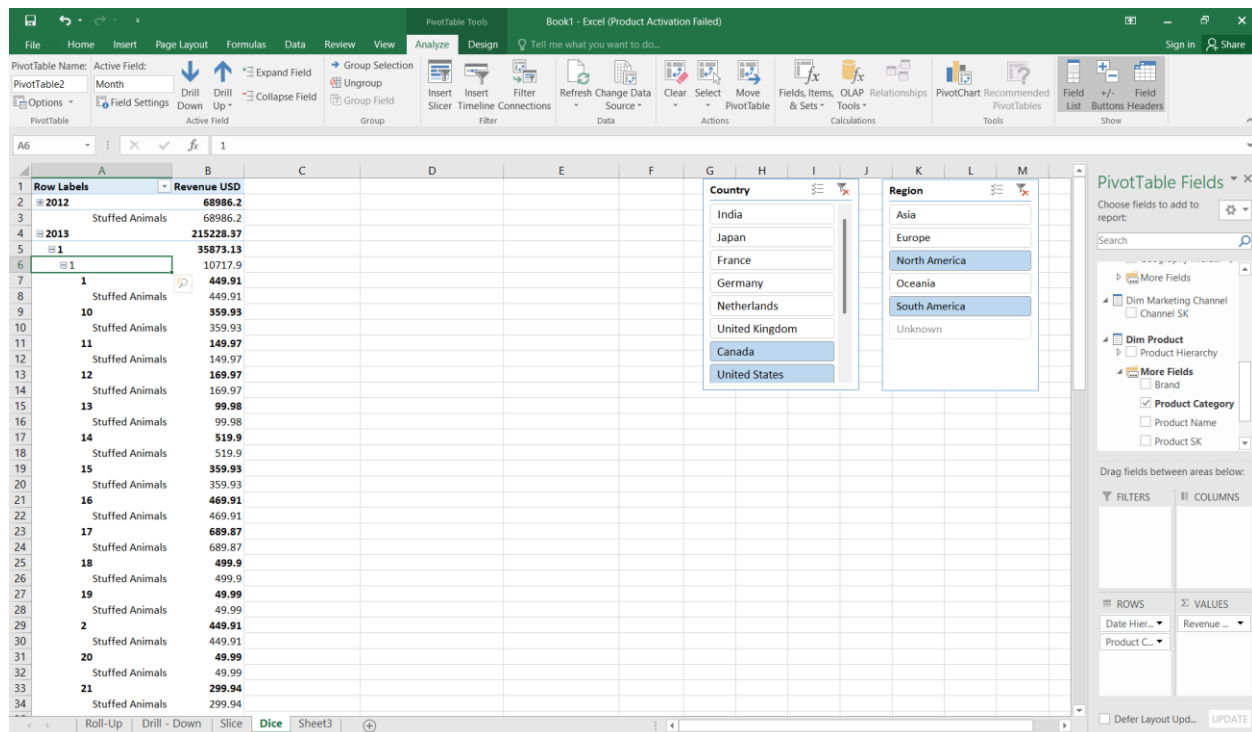


Figure 19: Dice operation — Data filtered by multiple dimensions (Country AND Region) simultaneously

### 3.6 Operation 5 – Pivot

Pivot is an OLAP operation that rotates the data cube to present the same data from a different analytical perspective. By swapping the row and column axes, the same underlying data is presented in a different orientation, enabling analysts to gain new insights without changing the actual data.

The Pivot operation demonstrates the ability to rotate the data cube to present the same data from a different analytical perspective. In View 1, the PivotTable is configured with Product Category as rows and Years (2012, 2013, 2014, 2015) as columns, allowing analysis of how each product category performed across different years. In View 2, the axes are rotated — Years become the rows and Product Category becomes the columns. This presents the exact same data but from a different perspective, now allowing analysis of how each year compared across different product categories. The Pivot operation does not change the underlying data — it only changes the orientation of how the data is presented, enabling analysts to gain different insights from the same dataset.

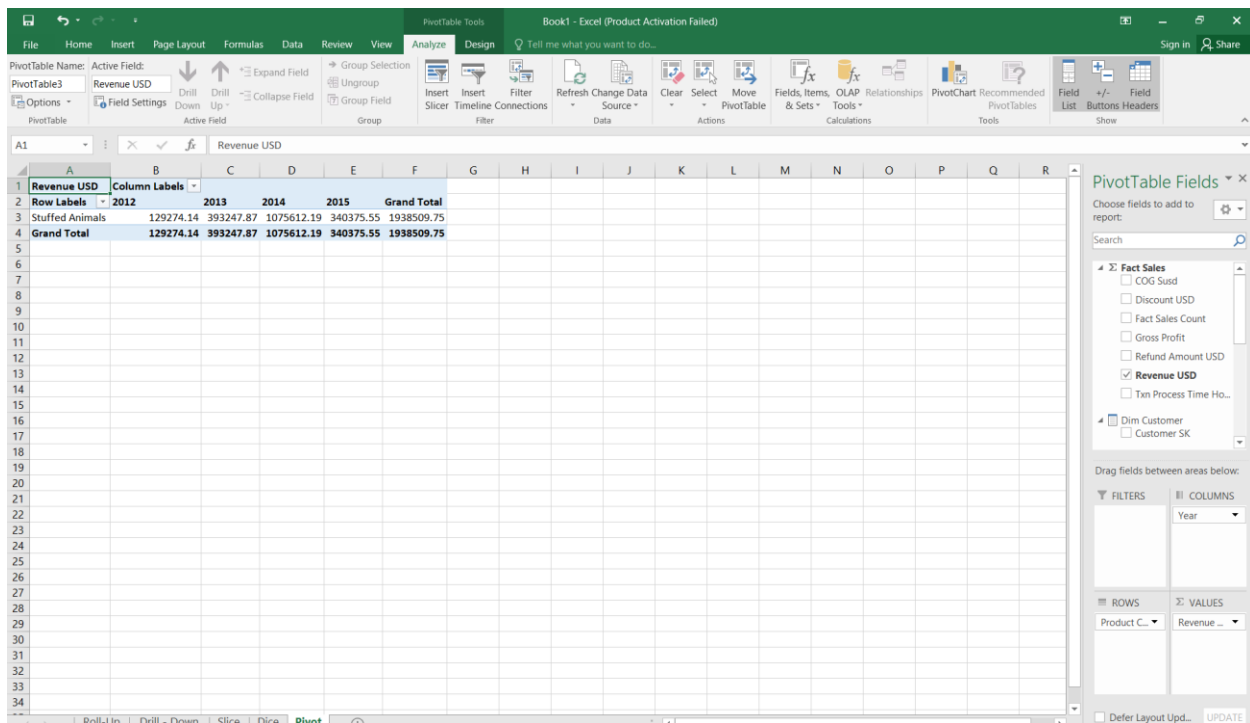


Figure 20: Pivot View 1 — Product Category as rows, Years as columns

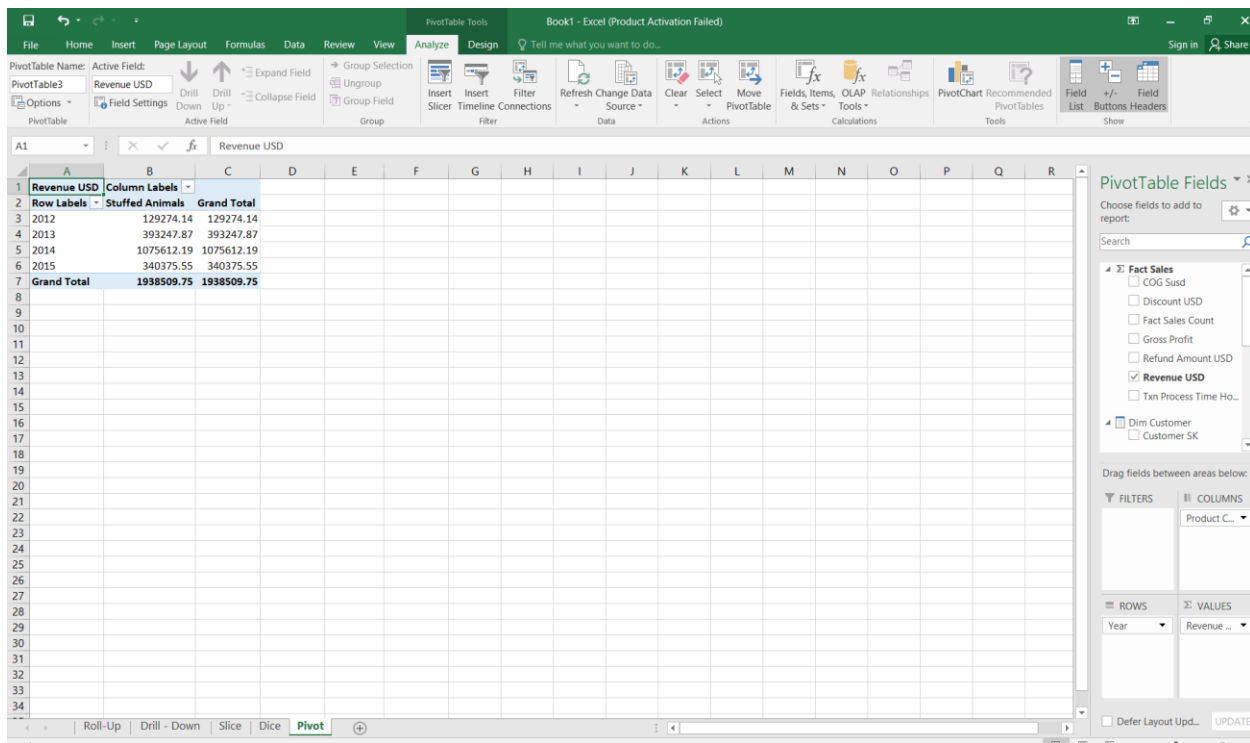


Figure 21: Pivot View 2 — Years as rows, Product Category as columns (axes rotated)

## Step 04: Power BI Reports

This step describes the development and publication of four Power BI reports built on the ToyStore\_DWH data warehouse. Power BI Desktop was connected to SQL Server using the server name LAPTOP-20K14P2C\MSSQLSERVER3 and all 7 tables were imported. All four reports were published to Power BI Service and demonstrated online at [app.powerbi.com](https://app.powerbi.com).

### 4.1 Data Preparation and Modeling

All 7 tables were loaded from ToyStore\_DWH into Power BI Desktop. The Model view was used to verify and confirm that all 6 relationships between FactSales and the dimension tables were correctly established. The relationships ensure that slicers and filters applied to dimension tables correctly filter the FactSales measures.

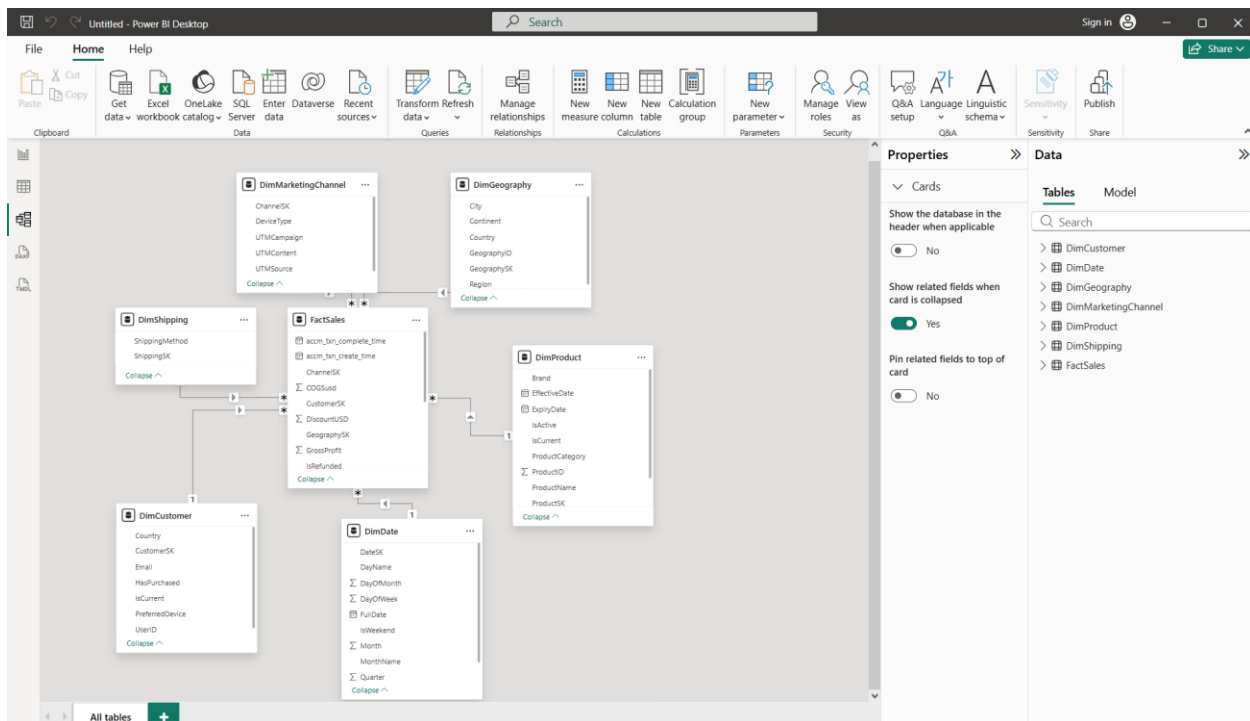


Figure 22: Power BI Model view showing all 6 relationships between FactSales and dimension tables

## 4.2 Report 1 – Matrix Visual

Report 1 demonstrates the use of a Matrix visual to display detailed tabular data with row and column groupings. The Matrix visual is similar to a PivotTable and allows hierarchical row and column organisation with multiple measures displayed simultaneously.

### Data Preparation

The Matrix was configured with DimProduct fields (Product Category and Product Name) as rows, DimDate Year as columns, and three measures from FactSales as values: Revenue USD, Gross Profit, and COGSUSD. This configuration allows users to analyse financial performance across different products and years in a single view.

### Visual Design

The Matrix visual was formatted with conditional background colour formatting applied to Revenue USD to highlight high-performing and low-performing products visually. Row headers were formatted for better readability. The report title was set to 'Sales Performance by Product and Year'. Two Card visuals were added showing Total Revenue USD and Total Gross Profit as summary KPIs.

Untitled - Power BI Desktop

Search

Sign in

FileHomeInsertModelingViewOptimizeHelp

PasteCutCopyFormat painterClipboard

Get dataExcel workbook catalogOneLake SQL ServerEnter dataDataaverseRecent sources

Transform dataRefresh dataQueries

New visualText box insertMore visuals

New visual calculationNew measureQuick measureCalculations

Sensitivity analysisSensitivityShare

PublishPrep data for CopilotAICopilot

Year

ProductCategory

| 2012              | 2013               | 2014           |                   |                    |                |                   |                    |                |
|-------------------|--------------------|----------------|-------------------|--------------------|----------------|-------------------|--------------------|----------------|
| Sum of RevenueUSD | Sum of GrossProfit | Sum of COGSUSD | Sum of RevenueUSD | Sum of GrossProfit | Sum of COGSUSD | Sum of RevenueUSD | Sum of GrossProfit | Sum of COGSUSD |
| 129,274.14        | 78,873.00          | 50,401.14      | 393,247.87        | 241,596.50         | 151,651.37     | 1,075,612.19      | 679,722.19         | 369,660.00     |
| 129,274.14        | 78,873.00          | 50,401.14      | 307,488.49        | 187,605.50         | 119,882.99     | 605,878.80        | 369,660.00         | 105,594.79     |
|                   |                    |                |                   |                    |                | 105,594.79        | 72,180.00          |                |
|                   |                    |                | 79,366.77         | 49,612.50          | 29,754.27      | 192,687.88        | 120,450.00         |                |
|                   |                    |                | 6,392.61          | 4,378.50           | 2,014.11       | 171,450.72        | 117,432.00         |                |
| 129,274.14        | 78,873.00          | 50,401.14      | 393,247.87        | 241,596.50         | 151,651.37     | 1,075,612.19      | 679,722.19         |                |

Filters

Search

Filters on this page

Add data fields here

Filters on all pages

Add data fields here

Visualizations

Build visual

Visuals

Py

Values

Add data fields here

Drill through

Cross-report

Keep all filters

Add drill-through fields here

Report 1 - Matrix

Figure 23: Report 1 — Matrix visual showing product performance with conditional formatting

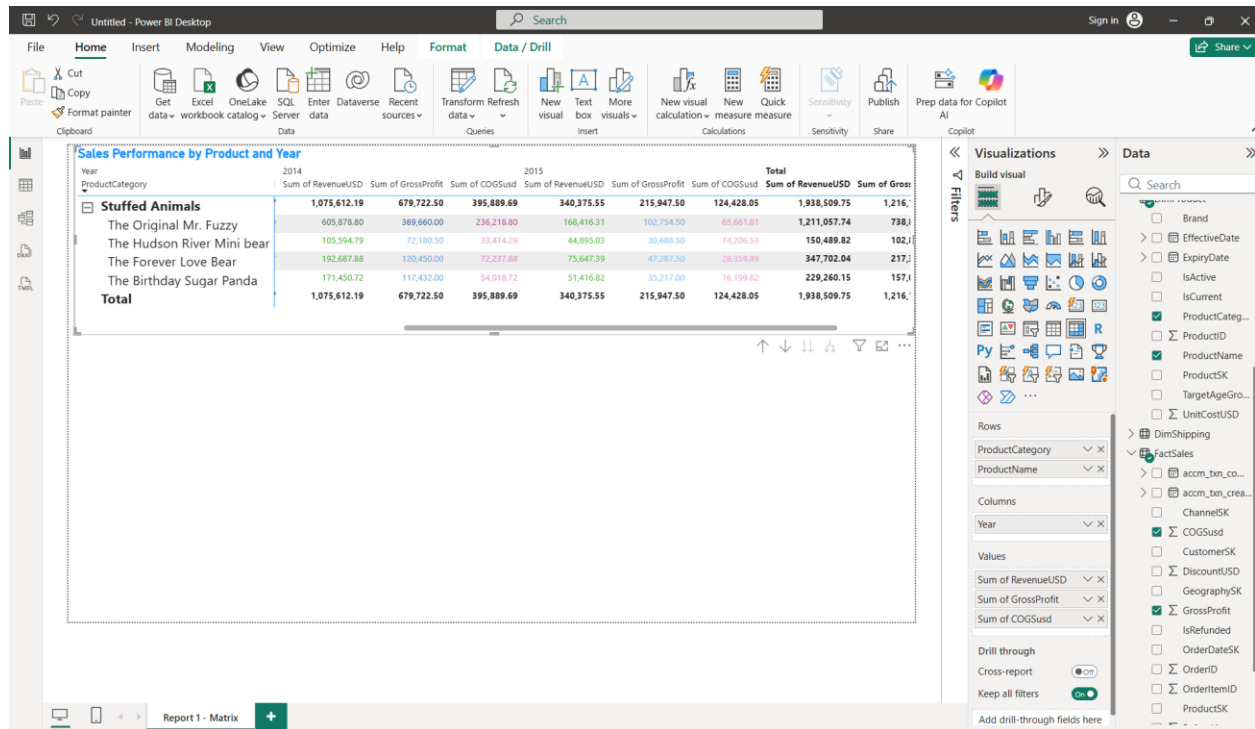


Figure 24: Report 1 — Matrix configuration showing rows, columns and values setup

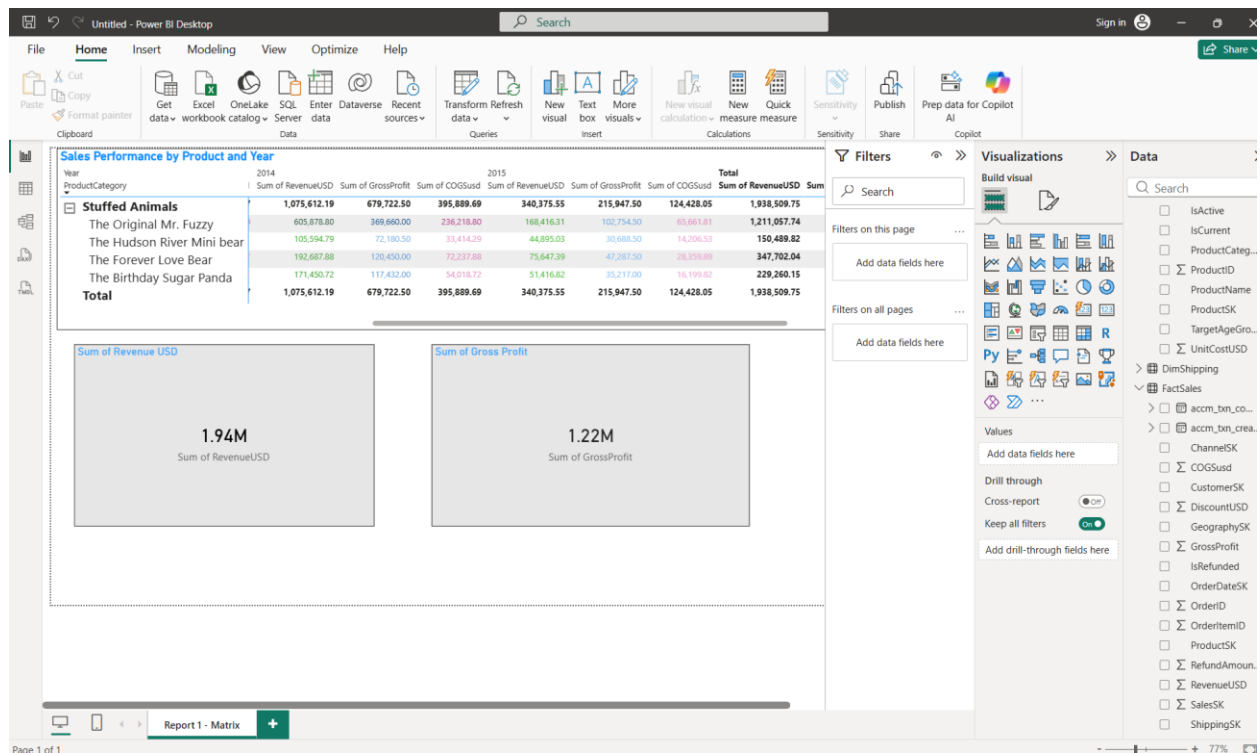


Figure 25: Report 1 — Complete view with Matrix and Card visuals showing Total Revenue and Gross Profit

### 4.3 Report 2 – Cascading Slicers with Multiple Charts

Report 2 demonstrates cascading slicers where selecting a value in the first slicer dynamically filters the options available in the second slicer. Three slicers and three charts were included to provide comprehensive interactive analysis of the data.

#### Cascading Slicers Implementation

Three slicers were added: Region (from DimGeography), Country (from DimGeography), and Shipping Method (from DimShipping). The Region and Country slicers cascade automatically because both fields belong to the same DimGeography table — selecting a Region filters the Country slicer to show only countries within that region. For example, selecting North America in the Region slicer automatically filters the Country slicer to show only United States and Canada.

#### Charts

Three charts were added to present insights: a Clustered Bar Chart showing Revenue by Product Category with Year as legend, a Line Chart showing Monthly Revenue Trend by Product



Category over time, and a Donut Chart showing Revenue distribution by Shipping Method. All three charts respond dynamically to slicer selections.

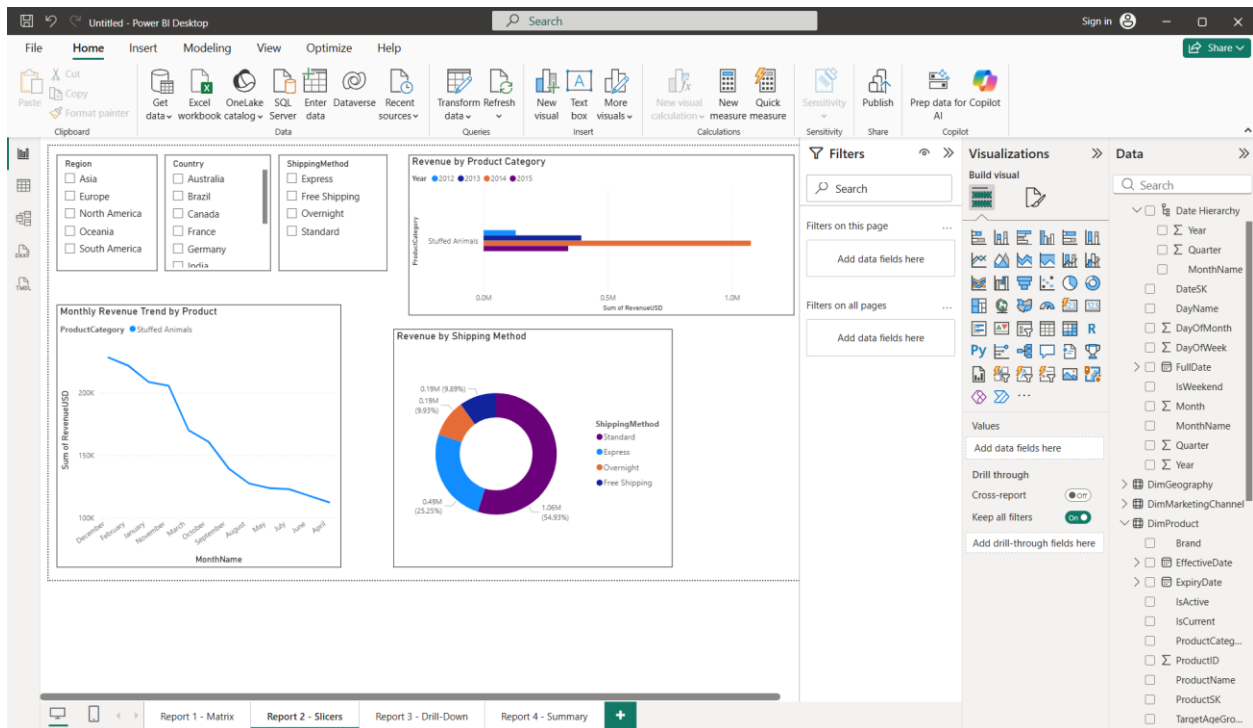


Figure 26: Report 2 — Three slicers and three charts showing cascading filter behaviour

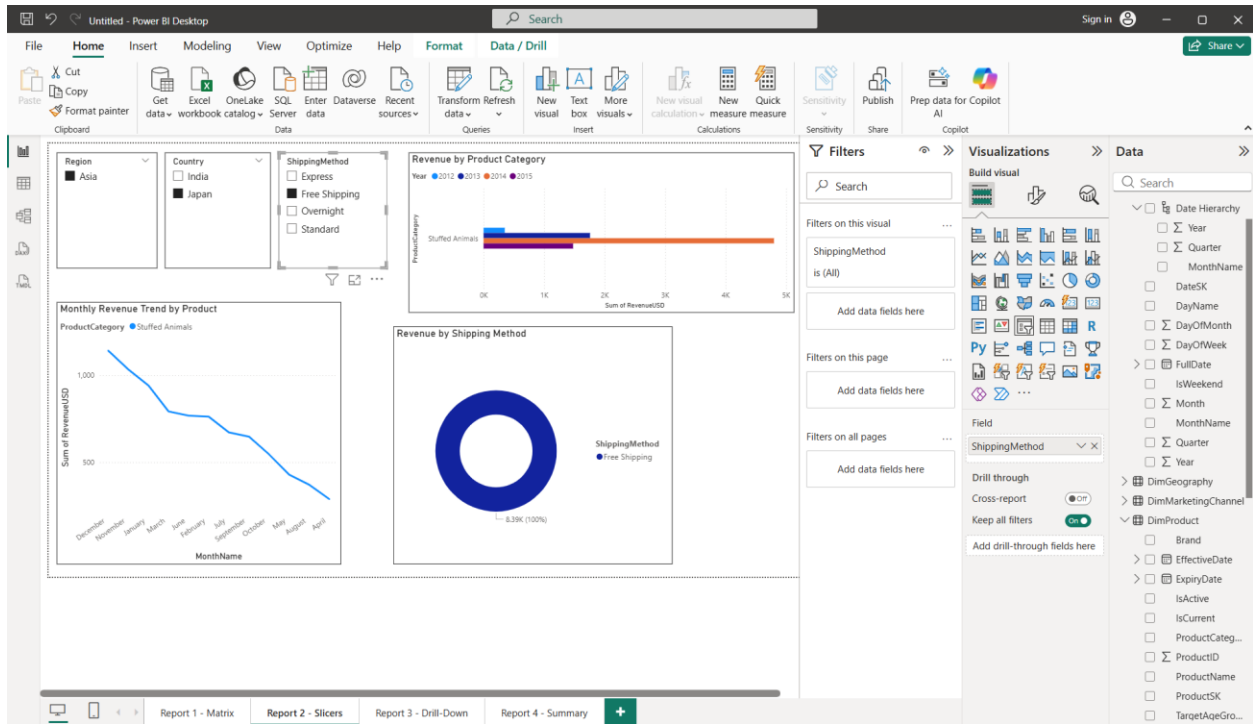


Figure 27: Report 2 — Cascading filter in action: selecting North America filters Country slicer dynamically

#### 4.4 Report 3 – Drill – Down Report

Report 3 demonstrates hierarchical drill-down navigation in Power BI charts. A Date Hierarchy (Year → Quarter → Month) was created in the DimDate table, enabling users to explore revenue data at progressively detailed time levels by clicking on chart elements.

##### Date Hierarchy Creation

A Date Hierarchy was created in the DimDate table in Data view by right-clicking Year and selecting New Hierarchy, then dragging Quarter and Month into the hierarchy. This hierarchy was then used as the X-axis of the Clustered Column Chart.

##### Drill-Down Charts

A Clustered Column Chart titled 'Revenue Drill-Down by Time' was created with Date Hierarchy on the X-axis and Revenue USD on the Y-axis. A Line Chart titled 'Gross Profit Drill-Down' was also added using the same Date Hierarchy. Both charts support hierarchical drill-down by clicking the drill-down arrow icon.

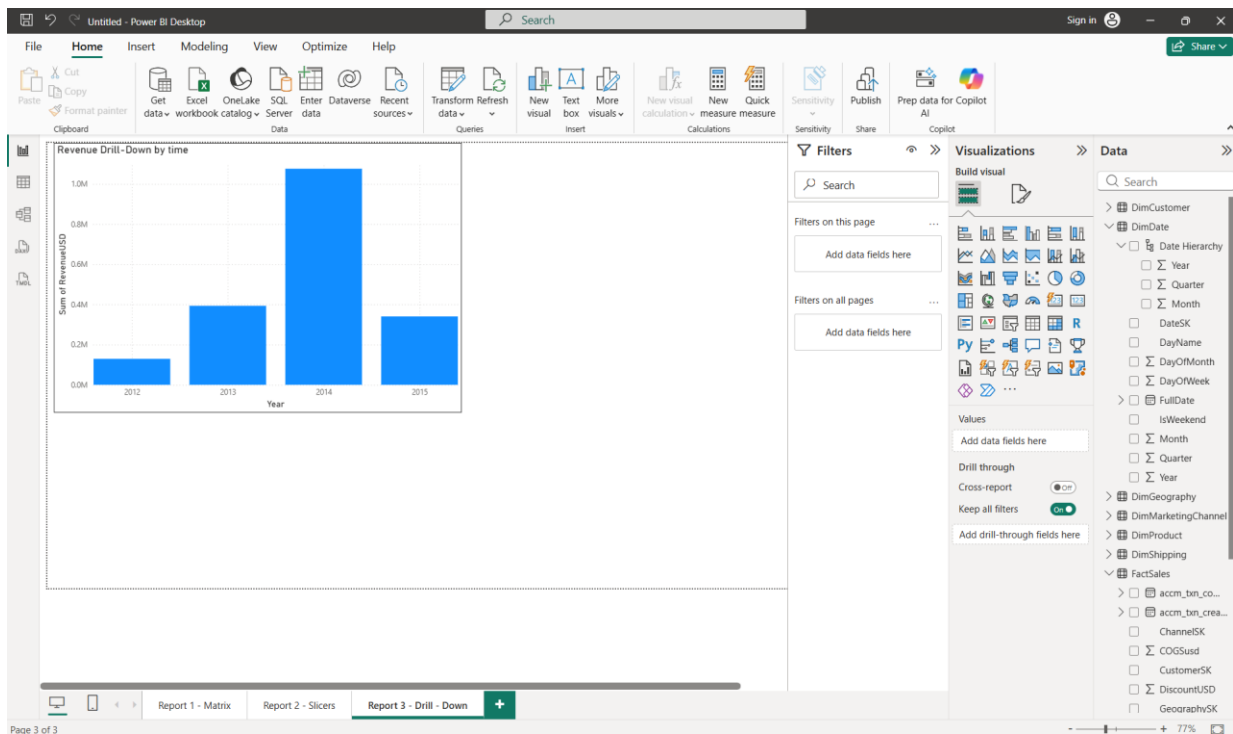


Figure 28: Report 3 — Column chart at Year level showing revenue for 2012, 2013, 2014, 2015

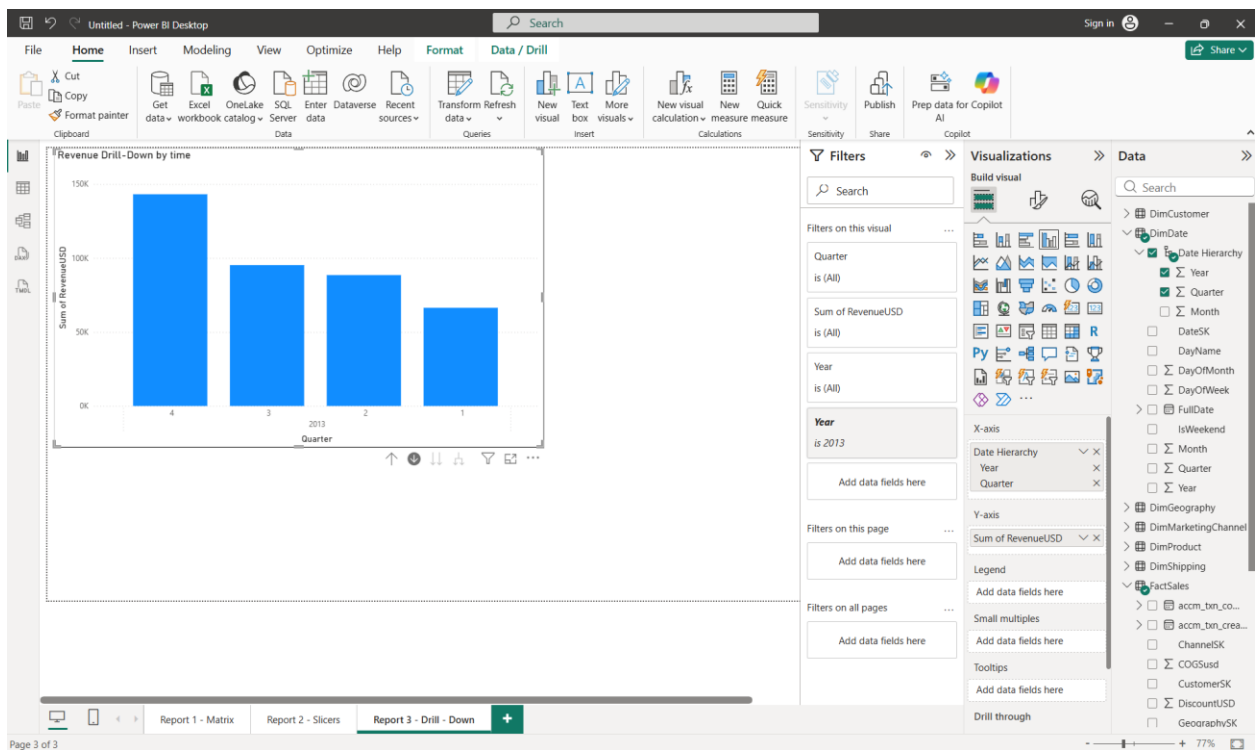


Figure 29: Report 3 — Drill-down to Quarter level showing Q1, Q2, Q3, Q4 of 2013

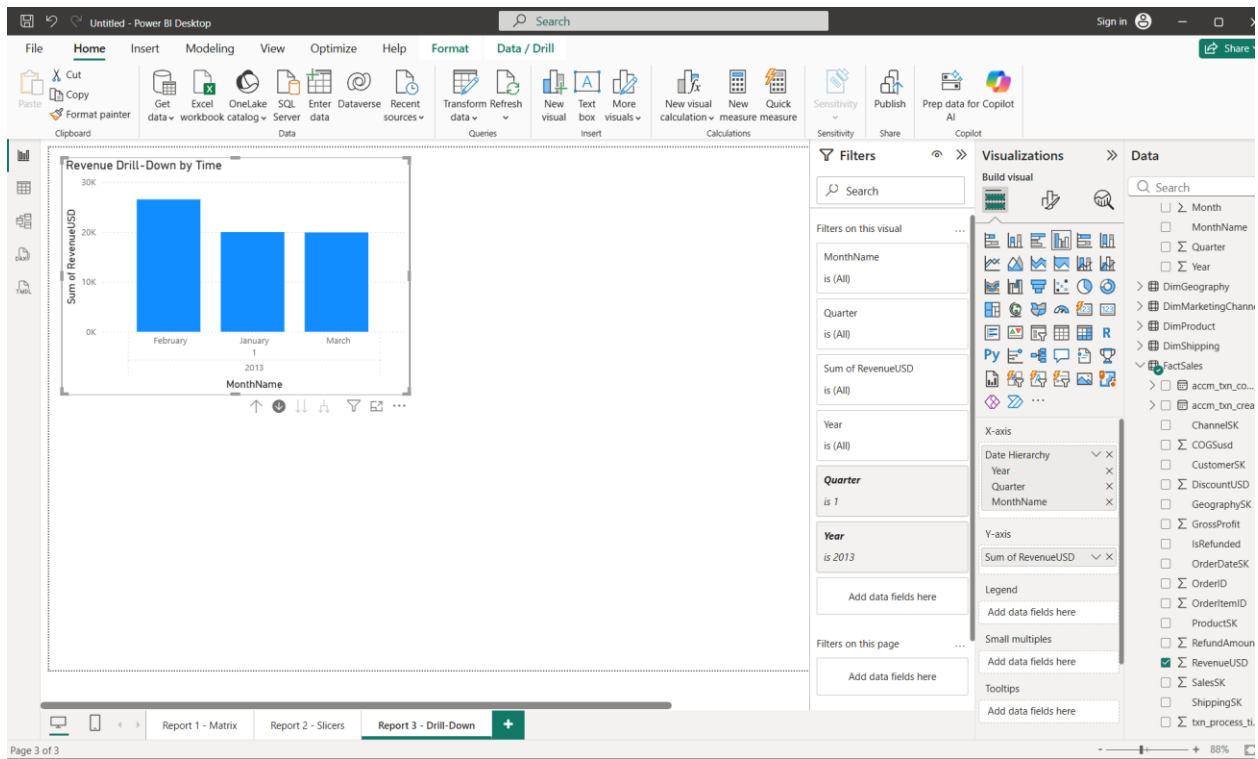


Figure 30: Report 3 — Drill-down to Month level showing individual monthly revenue

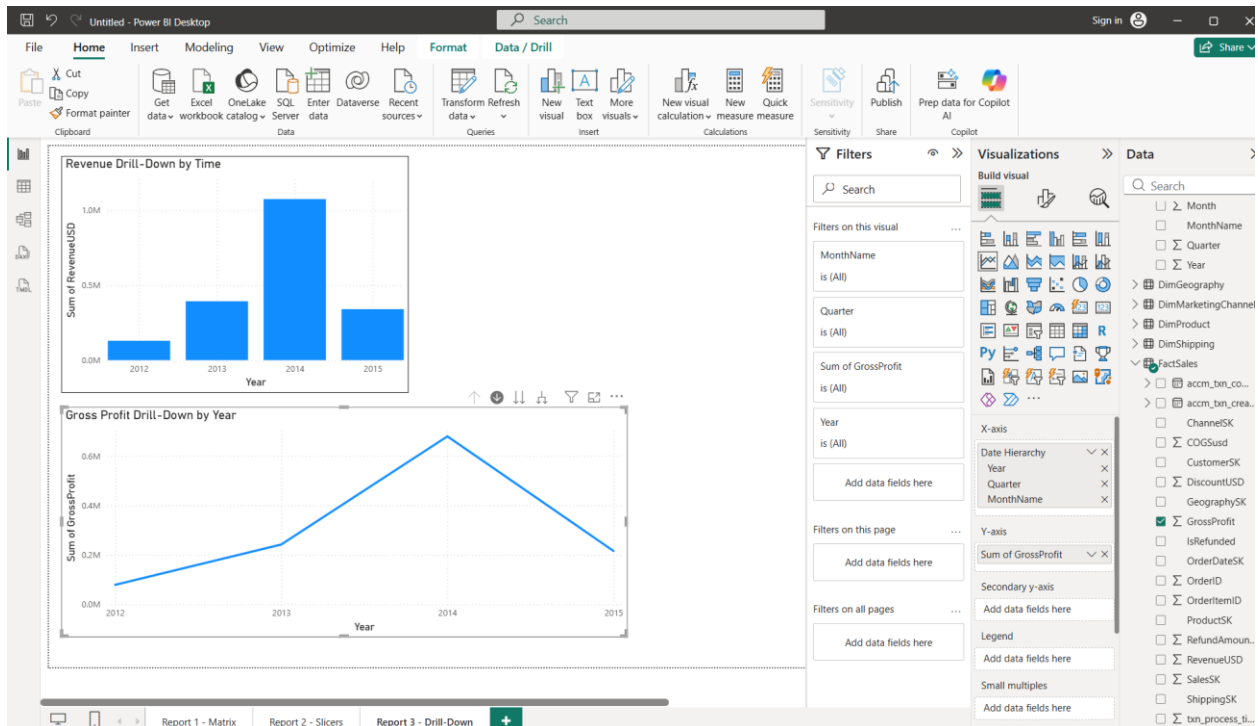


Figure 31: Report 3 — Line chart showing Gross Profit drill-down at Year level

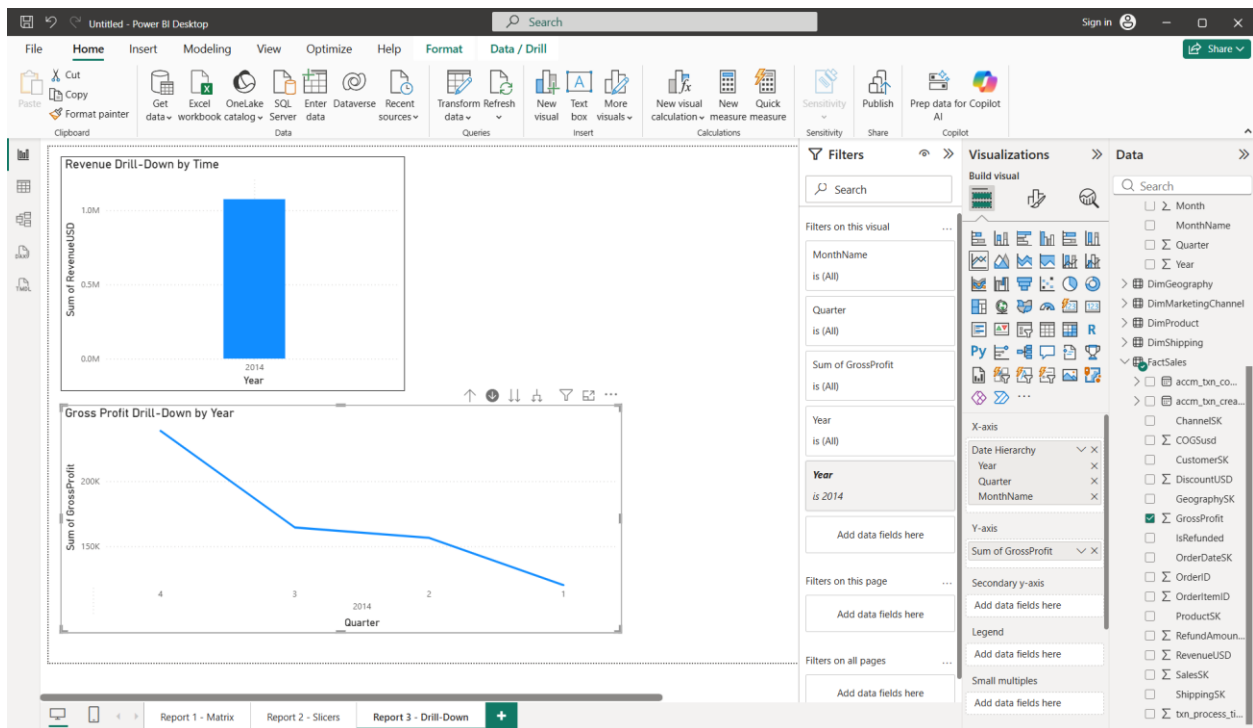


Figure 32: Report 3 — Line chart drilled down to Quarter level

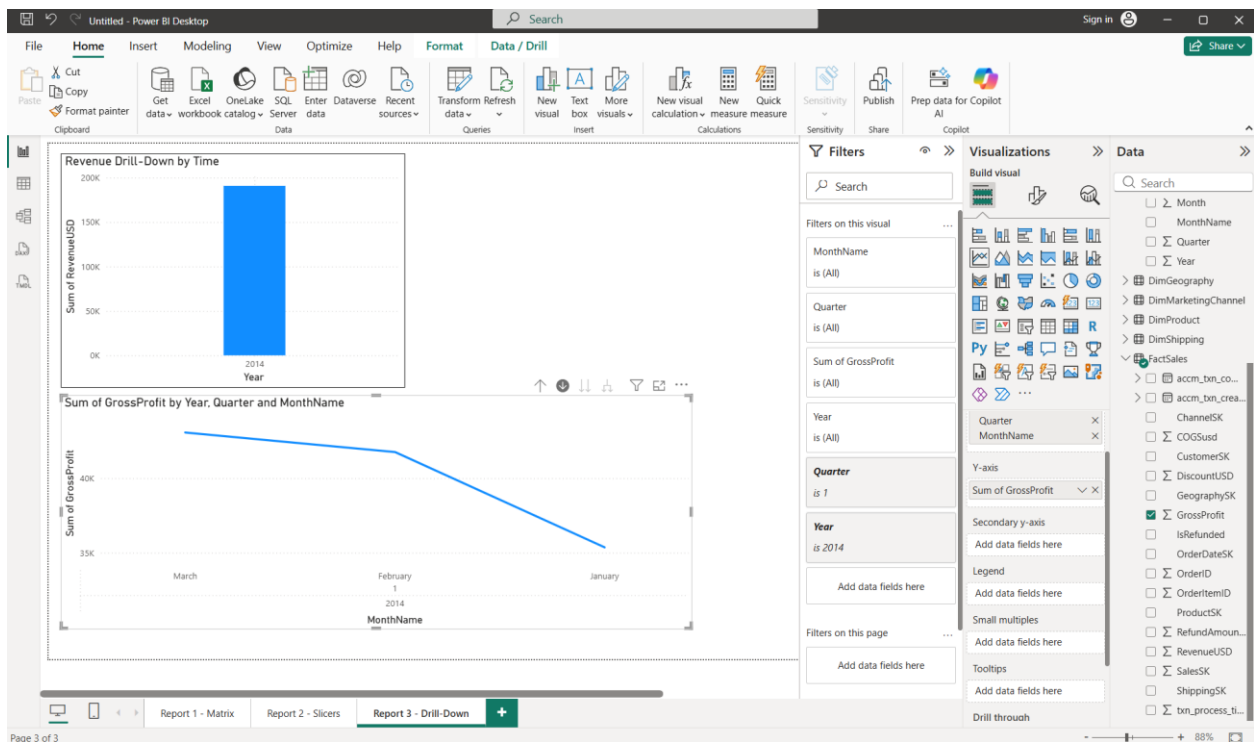


Figure 33: Report 3 — Line chart drilled down to Month level

## 4.5 Report 4 – Drill – Through Report

Report 4 demonstrates the drill-through feature in Power BI, which allows users to right-click on a data point in a summary visual and navigate to a detailed page filtered for that specific data point. Report 4 consists of two pages: a Summary page and a Detail page.

### Summary page (Report 4 – Summary)

The Summary page contains a Bar Chart showing Revenue by Product Category and a Map visual showing Revenue by Country. This page gives users a high-level overview of performance across all product categories and geographic regions.

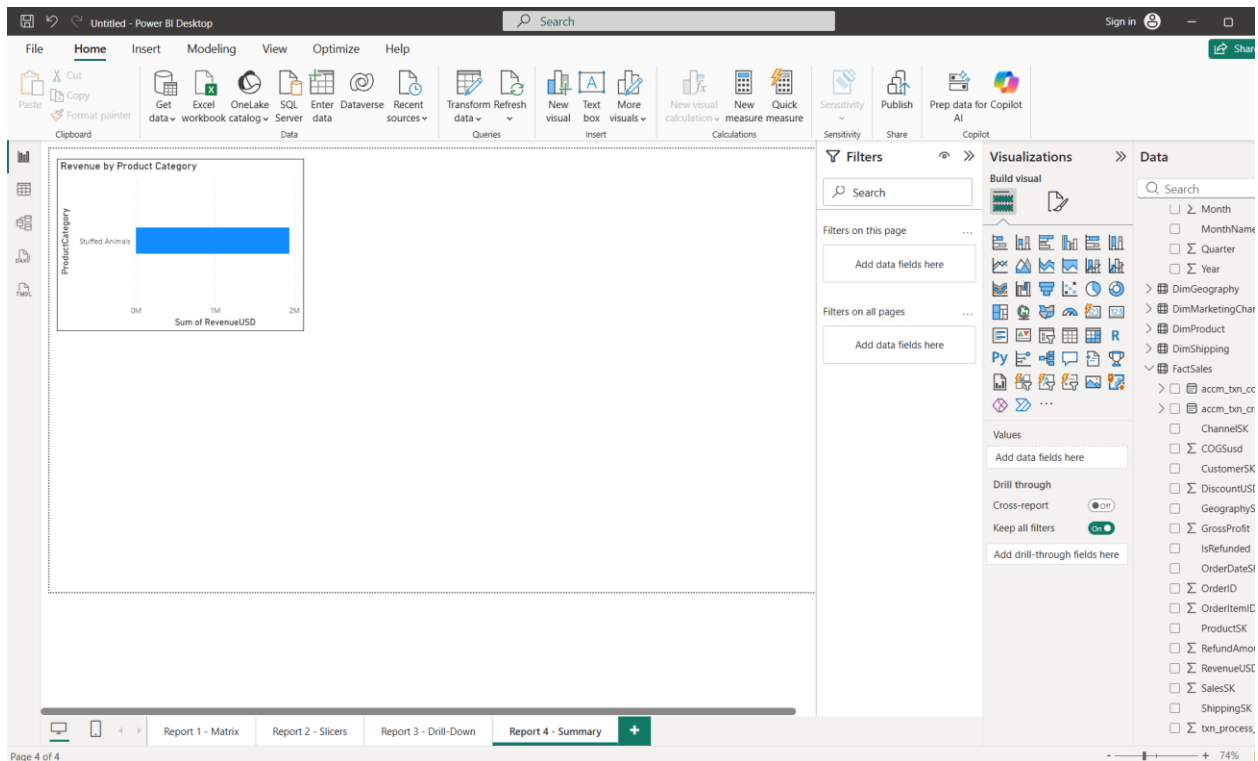


Figure 34: Report 4 Summary page — Bar chart and Map showing revenue by product category and country

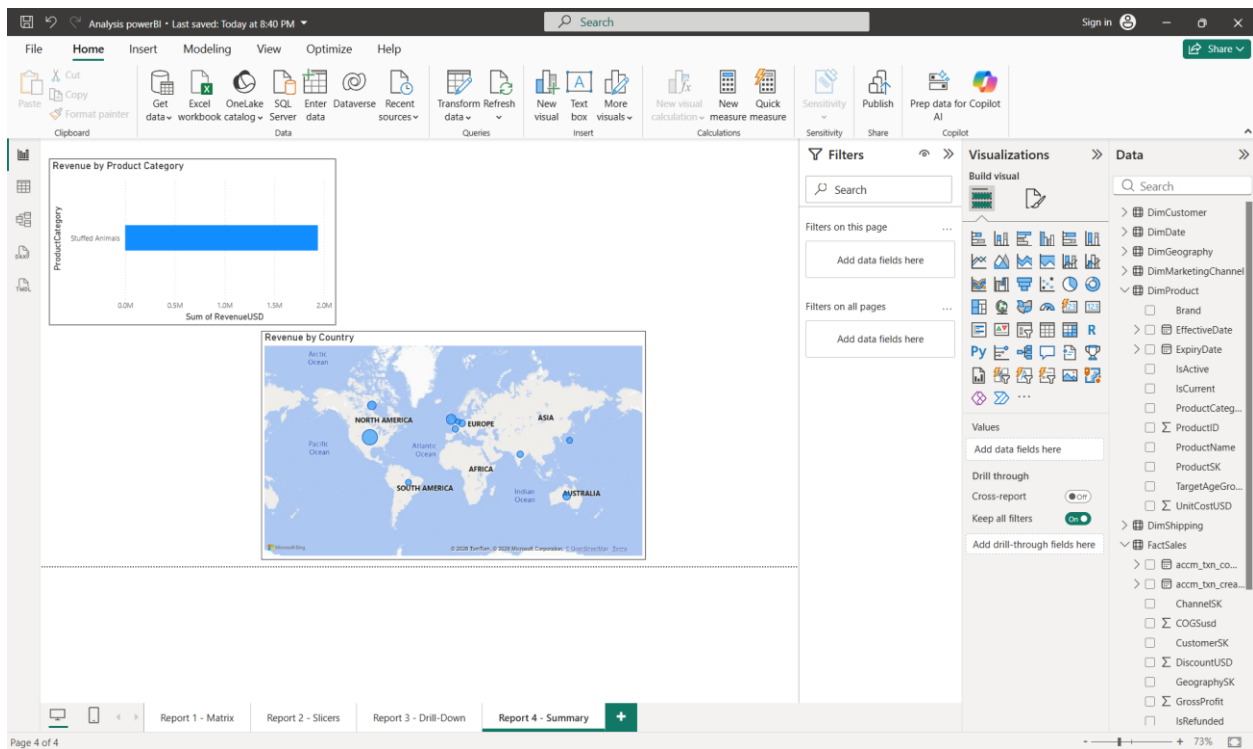


Figure 35: Report 4 — Right-click menu showing Drill through option to navigate to Detail page

## Detail Page (Report 4 – Detail)

The Detail page was configured with DimProduct → Product Category as the Drill Through field. This means when a user right-clicks on any product category in the Summary page and selects Drill Through → Report 4 - Detail, the Detail page opens automatically filtered to show only data for that selected category. The Detail page contains a Table visual showing product-level details, a Line Chart showing monthly revenue trend, and a Card showing total revenue for the selected category.

## Drill-Through Setup

The Drill Through field was set by dragging Product Category from DimProduct into the Drill through section in the Visualizations pane on the Detail page. Power BI automatically adds a back navigation button to allow users to return to the Summary page after viewing the details.

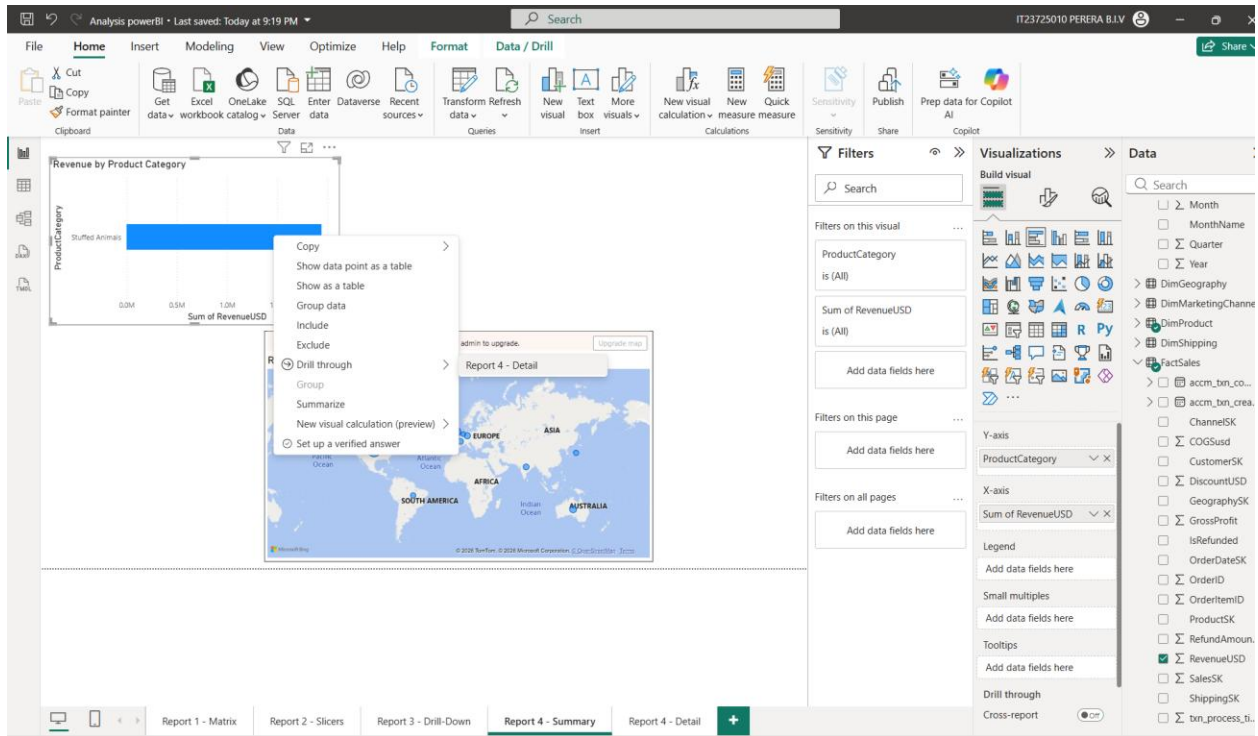


Figure 36: Report 4 Detail page setup showing drill-through configuration

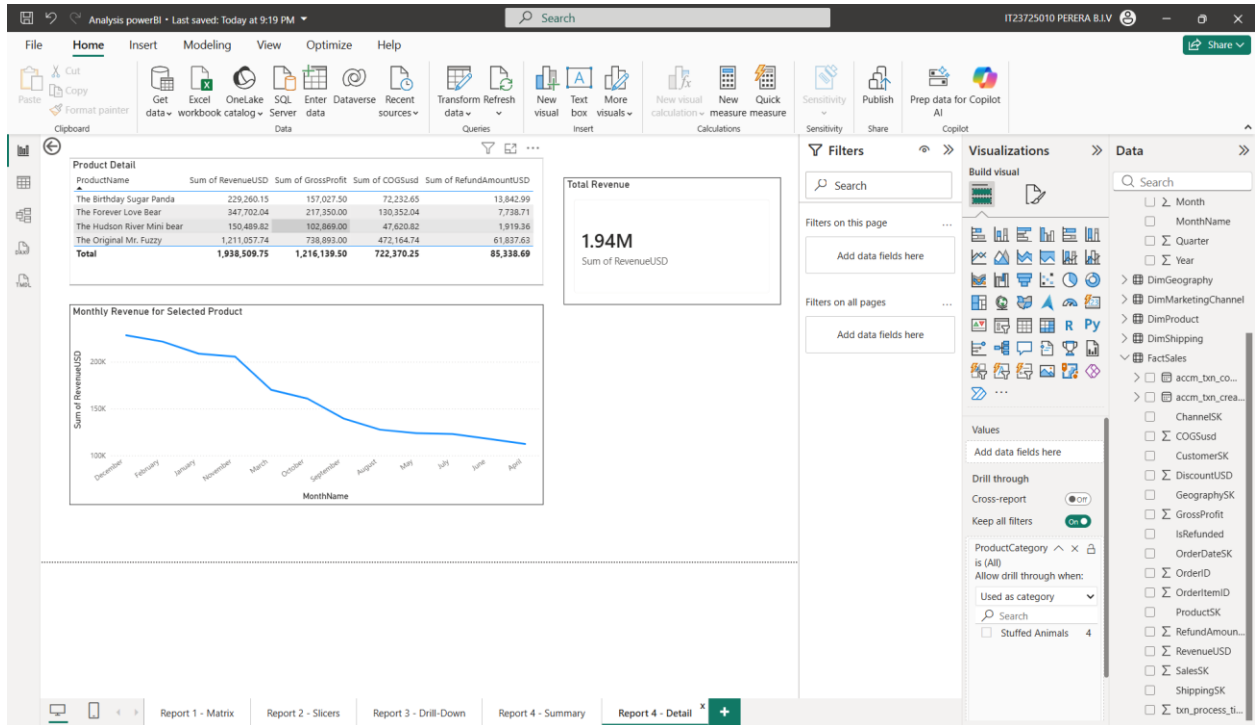


Figure 37: Report 4 Detail page filtered to show data for selected product category



## 4.6 Publish to Power BI Service

All four reports were published to Power BI Service using the Publish button in Power BI Desktop. The reports were published to My Workspace and are accessible online at [app.powerbi.com](https://app.powerbi.com). The following screenshots demonstrate each report running in the Power BI Service browser, confirming successful online publication.

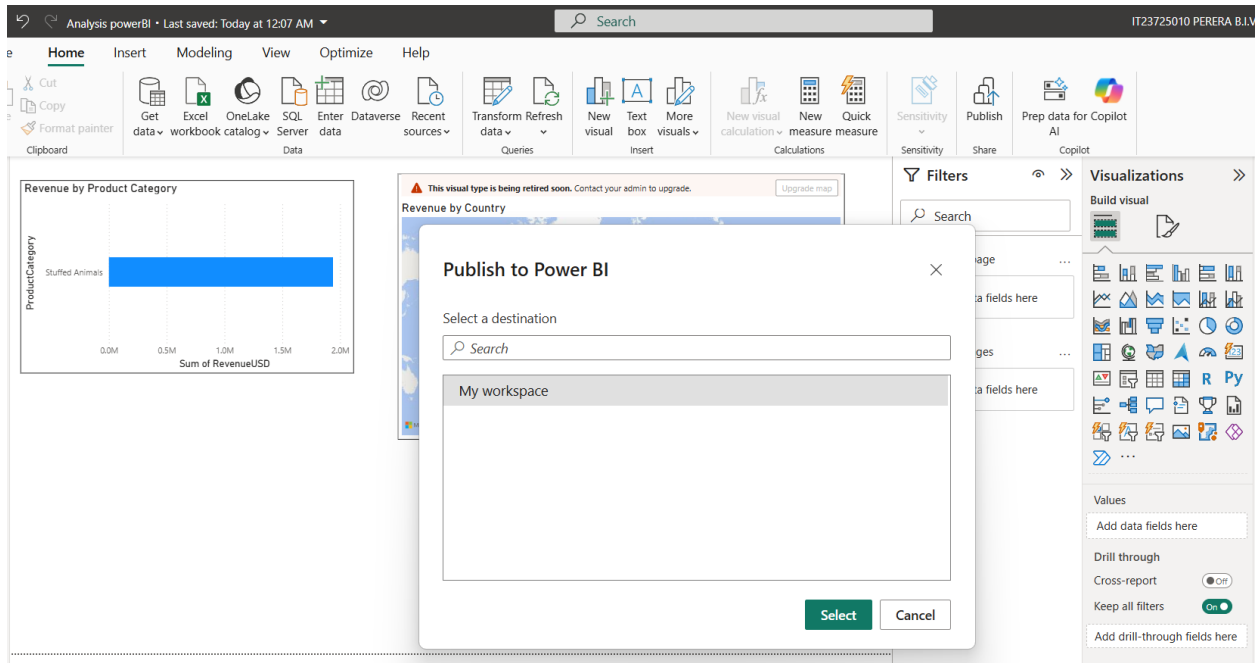


Figure 38: Publishing to Power BI Service — successful publication confirmation

Figure 39: Report 1 — Matrix Visual open in Power BI Service at [app.powerbi.com](https://app.powerbi.com)

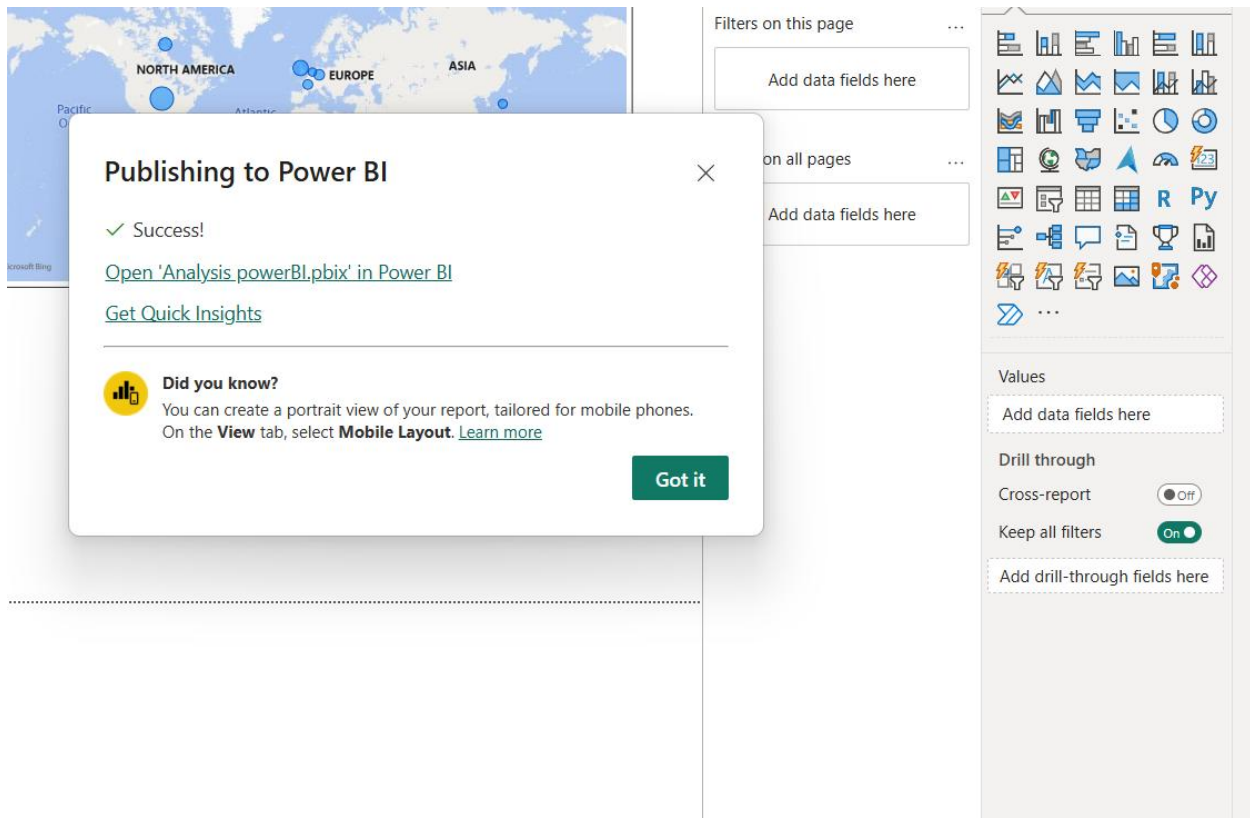


Figure 39: Report 1 — Matrix Visual open in Power BI Service at app.powerbi.com

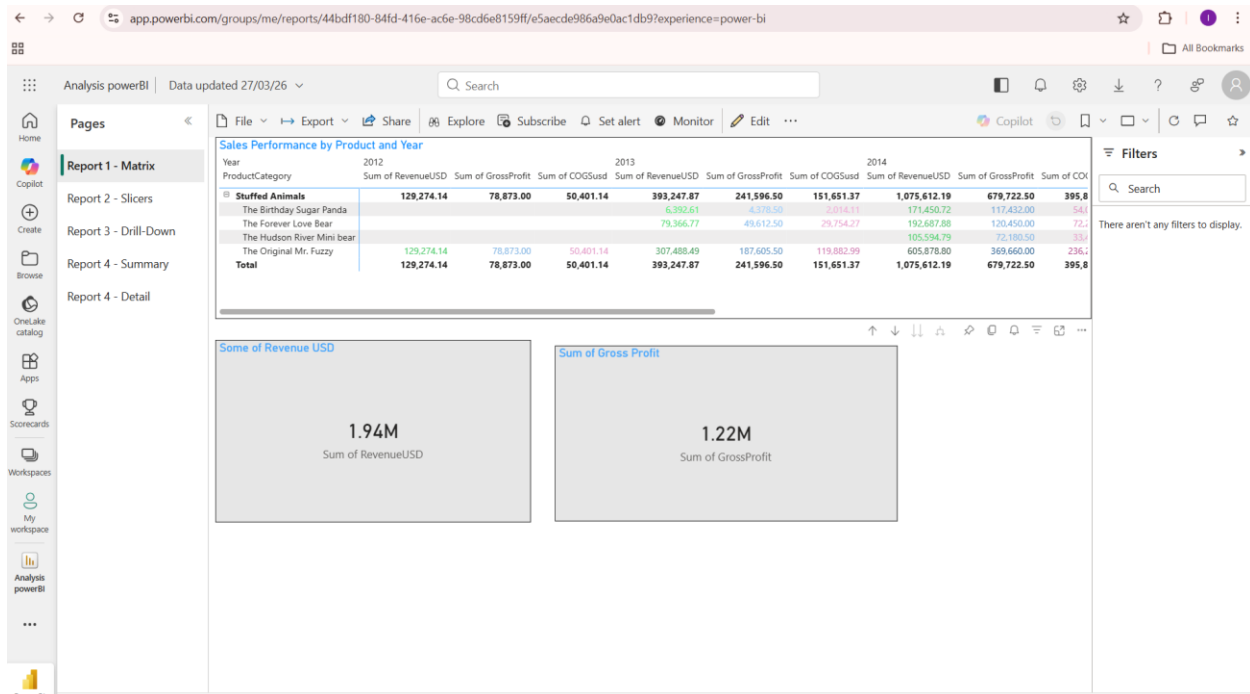


Figure 40: Report 2 — Cascading Slicers open in Power BI Service

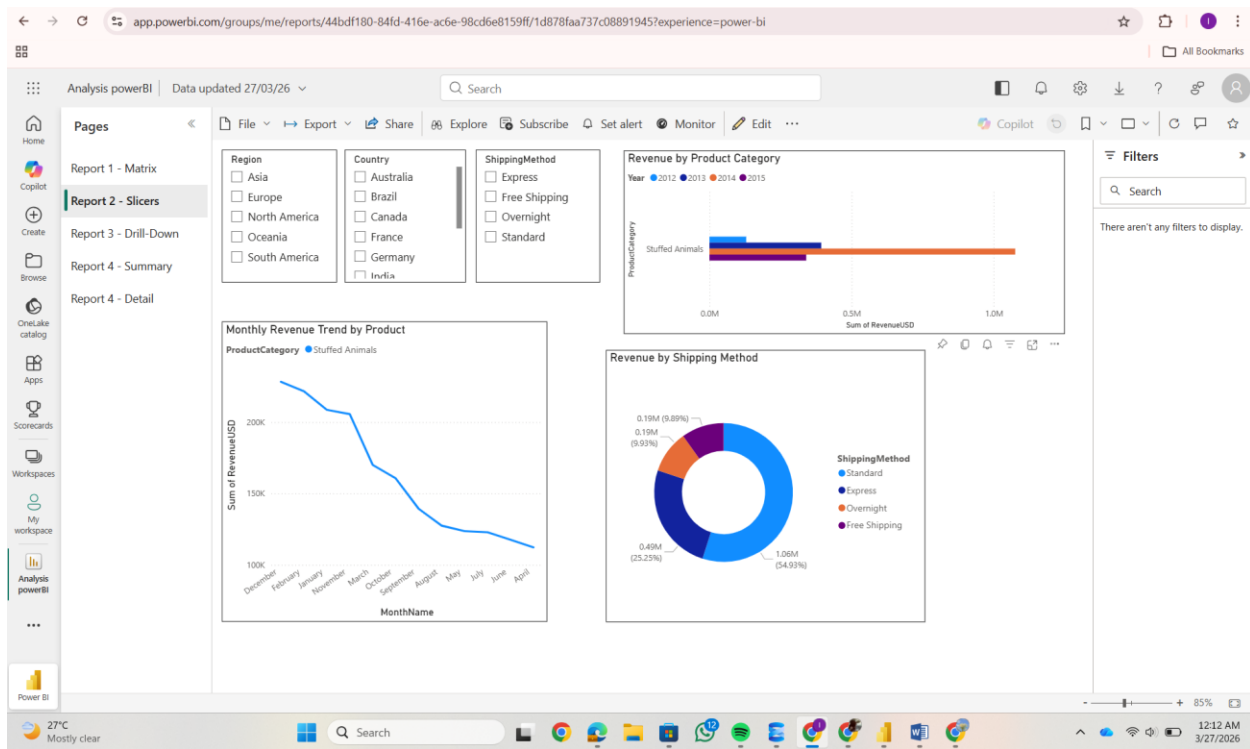


Figure 41: Report 3 — Drill-Down Report open in Power BI Service

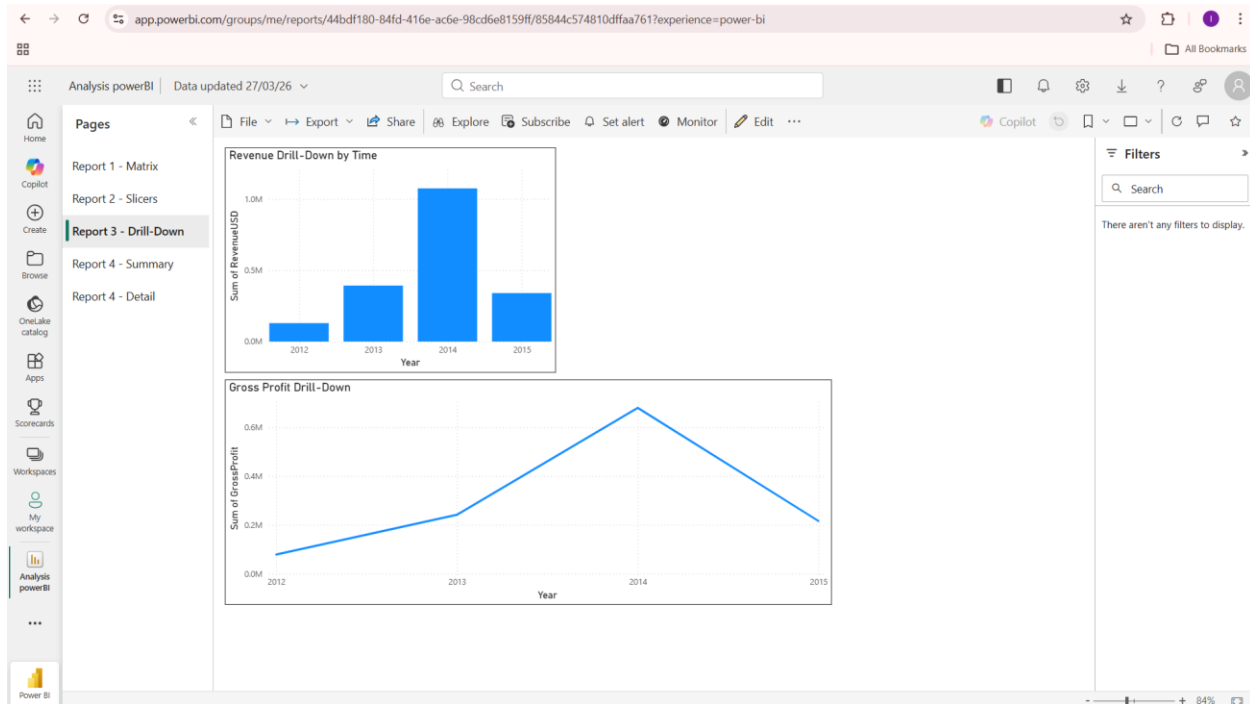


Figure 42: Report 4 Summary page open in Power BI Service

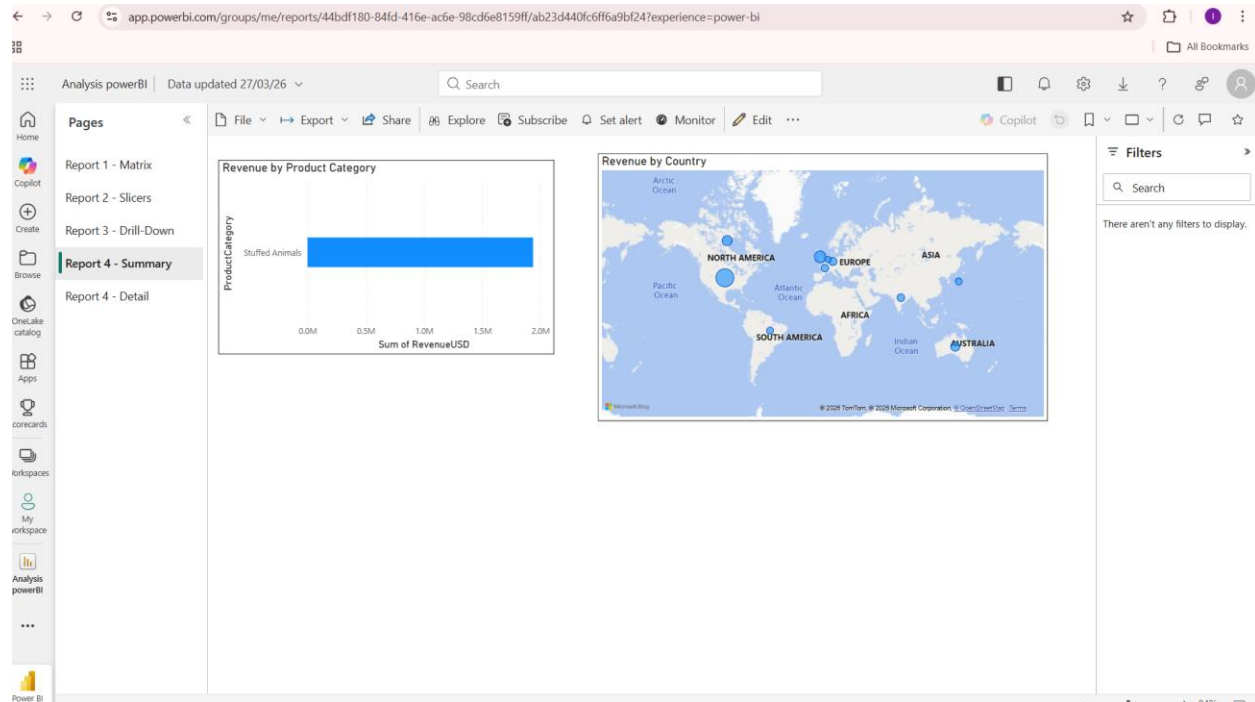


Figure 43: Report 4 Detail page open in Power BI Service

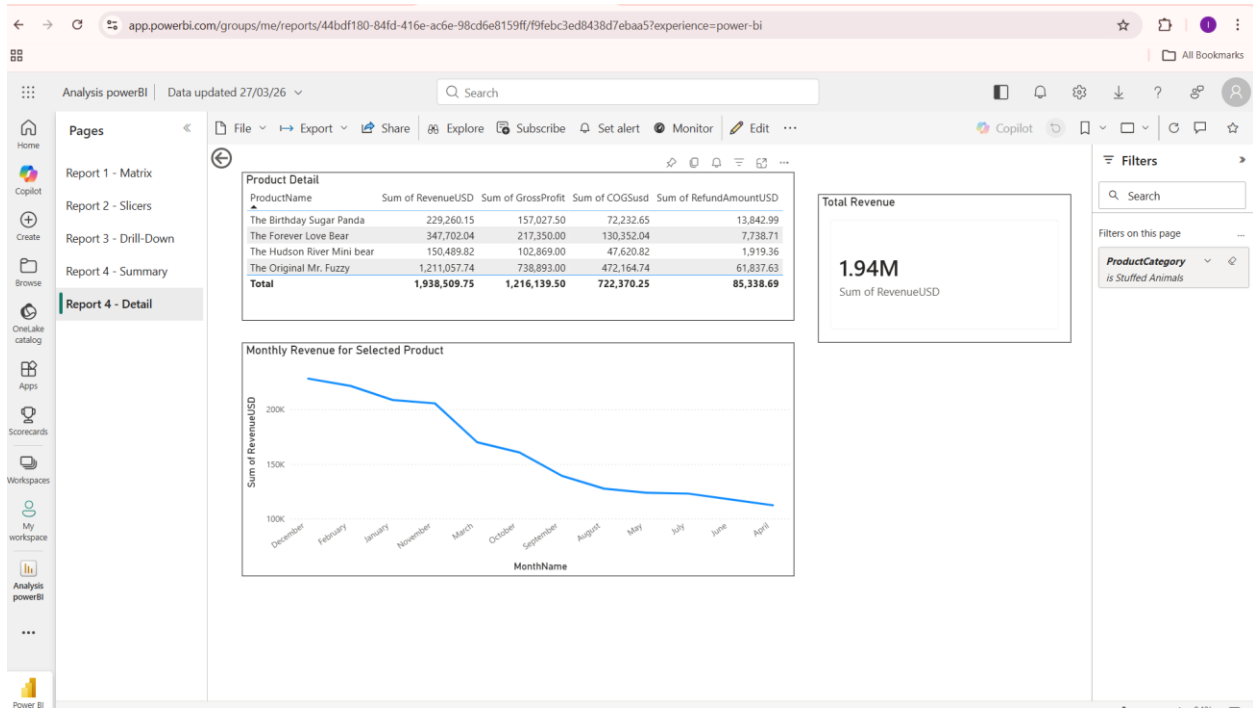


Figure 44: All reports visible in Power BI Service workspace